

**Pervasive and User Focused Biometrics Border Project
(PROTECT)
H2020 – 700259**

D6.5 Specification of new enhanced electronic passport for PROTECT

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2018-02-26	0.9	Frank Schmalz	Removed confidential content
2018-10-10	1.0	Frank Schmalz	Update of Abbreviations
2018-10-30	1.1	Frank Schmalz	Update of Section 2 to explain the purpose of the certificates.
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2018-11-09	1.3	Frank Schmalz	References to the underlying specifications in Sections 3 and 4 have been inserted. To avoid redundancy this information has not been copied into this document but referenced. The ICAO LDS2 specification is now publicly available (see references). The BSI and LDS2 specifications specify all command and file structures.

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Abbreviations

BAC	Basic Access Control
CVCA	Card Verifiable Certificate
Data_page	The page of a passport booklet containing the machine-readable zone, the passport holder photo and his biographical data.
DG	Datagroup
DV	Document Verifier
EAC	Extended Access Control
eMRTD	electronic Machine-Readable Travel Document. Also the name of the standard electronic passport application on the chip.
ICAO	International Civil Aviation Organization
IFD	Interface Device. The electronic passport reader.
LDS	Logical Data Structure. This is the storage structure for data on electronic passports. It is specified in ICAO Doc9303 [5]
LDS2	Logical Data Structure 2. Version 2 of LDS. At the time of writing still draft. Specified in [7]
MRTD	Machine-Readable Travel Document
MRZ	Machine readable zone
PACE	Password Authenticated Connection Establishment
PICC	Proximity Integrated Circuit Card
EF	Elementary File. A container to store data on the smartcard. Usually and especially with ePassports this is implemented according to ISO 7816-4. See [10] page 19
DF	Dedicated File. A container with files like a directory or folder on PC operating systems. DFs are implemented different on javacard and native cards. On javacard each DF is a separate applet with firewall protection. For more details see [10] page 19

1 Introduction

After describing different possibilities to achieve the goals of the PROTECT project concerning enhanced electronic passports in deliverable D6.4 the next task has been to implement selected ideas from the report.

The report concludes that:

- “The objective of “...a most fluent non-intrusive control process...” is the hardest to achieve without giving up any data protection or privacy properties of the current system.”
- “For the secure storage of additional biometrics 2 promising approaches have been shown. With Logical Data Structure 2 being the one with the least ethical implications as well as same or higher level of security as the current solution”

Changing specifications of electronic passports is a lengthy process that requires consent of all members of the ICAO new technology working group. Logical Data Structure 2 is currently specified and will be approved soon. All other approaches to allow the storage of additional biometrics discussed in Deliverable D6.4 have serious implications on security. Members of the ICAO NTWG have rejected these proposals for this reason.

Implementing a passport that is not compliant to ICAO LDS specifications will basically make exploitation impossible. Such a passport will not be accepted by issuing or verifying states. A standardized security evaluation will not be possible due to the lack of independent test specifications and protection profiles.

Therefore the most promising approach is the implementation of a LDS2 compliant passport for protect and evaluate its usability for storing additional biometrics and supporting the PROTECT scenarios.

1.1 Purpose of the document

According to the “Description of the action” [3] the purpose of this report is to specify “the logical data structure 2 electronic passport application” which is used to store additional biometrics.

Section 2 lists requirements from D3.1 and their impact on the specification. However most requirements collected in Deliverable D3.1 do not relate to the electronic passport. Therefore the major requirements are coming from the LDS2 specification [11] and have been taken into account.

Section 3 describes the data structures especially the certificates necessary to read and write additional biometrics to the electronic passport.

Section 4 describes the actual applet implementation with details on the supported protocols and how the necessary data is stored on the security chip. The ICAO specification does not give details on a specific implementation but merely specifies the interfaces to this electronic passport. The main work in the PROTECT project concerning this document has been undertaken in specifying and developing a working ePassport app that supports this interface. For a javacard this is no trivial case since the first electronic passport applications have been designed with a single ISO7816-4 application in mind. The multiple DFs result in multiple applets with inter applet communication sharing security states. There is also no specification how administrative data is stored on the chip, how the applets are installed on the chip, or how the passport is personalized. Section 4 addresses these issues.

1.2 Outcome

The major theoretical/academic work in PROTECT concerning the advanced electronic passport has been contributed to the Deliverable D6.4 and the standardization work in the ISO/IEC JTC 1 / SC 17 / WG3 "Machine Readable Travel Documents" leading to the final version of the Logical Data Structure 2 specification [11]. The major implementing work has been performed by Veridos's linked third party G+D in implementing reference javacard applications according to this specification. The javacard applications can now be used to demonstrate a possible usage scenario with classic travel documents according to ICAO.

The scenario will most likely be superior to the mobile scenario in terms of security, compatibility and privacy but inferior in ease of use and speed. The limited storage space of security chips might lead to problems if template sizes of additional biometrics are too high.

2 General Requirements

Logical Data Structure 2 is a specification generated by the ICAO New Technology Working Group and the ISO/IEC JTC 1 / SC 17 / WG3 "Machine Readable Travel Documents". The specification introduced 3 new data containers on electronic passports:

1. Electronic travel stamps
2. Electronic Visas
3. Additional biometrics

PROTECT is attempting to use this new specification to store the templates of additional biometric researched in the project on an ICAO compliant 4th generation passport.

2.1 Requirements from D3.1 User requirements and scenarios

2.1.1 Set A – General Requirements

Req-ID	Description	Status
GEN_001	The PROTECT system demonstrations must be able to be inserted into operational or near-real environments without undue disruption to everyday activities.	Not applicable to the electronic passport application
GEN_002	The PROTECT system demonstration scope must not impact upon classified border control databases or other sensitive operations performed by border guards.	Not applicable to the electronic passport application
GEN_003	The PROTECT system demonstrations must be able to simulate normal border control operations, where integration into existing systems is not possible: e.g. border control databases checking (National, SIS II, Interpol, etc.) and enable easy identification of not eligible travellers.	Not applicable to the electronic passport application
GEN_004	The PROTECT system demonstrations must include software and hardware systems to measure performance and transaction times	Performance will depend on the size of the templates provided by the partners. Performance can be measured when using the 4 th gen. passport
GEN_005	The PROTECT system demonstrations must demonstrate innovative methods of traveller identification and assessment – including single/multi/fused modality biometric capture and recognition.	Not applicable to the electronic passport application
GEN_006	The system must include devices to collect machine-readable zone (MRZ) and RFID chip data from eMRTDs which are simple to operate.	Not applicable to the electronic passport application

GEN_007	The system, as demonstrated in the location, must comply with local health and safety regulations and standards.	Not applicable to the electronic passport application.
GEN_008	The PROTECT demonstrations must comply with the European General Data Protection Regulation (GDPR) principles where personal data is being captured and processed.	The app is running on a security chip. Biometric data added by PROTECT is protected by the EAC protocol.
GEN_009	The PROTECT demonstrations demonstrate respect for human dignity and not discriminate against individuals on grounds unrelated to lawful border control.	Not applicable to the electronic passport application.

2.1.2 Set B – Operational Requirements – Passport and Biometric Capture

Req-ID	Description	Status
OP_001	The BCA must be a designated area with physical and secure boundaries which is placed in an appropriate place on the designated route through the in- or outbound control zone.	Not applicable to the electronic passport application..
OP_002	Traveller flow through the BCA should be in one direction only.	Not applicable to the electronic passport application.
OP_003	The enclosure surfaces around and within the BCA must be transparent	Not applicable to the electronic passport application.
OP_004	Travellers walking through the BCA may be streamed into separate lanes either by physical barriers or markings so that they proceed in more or less single file.	Not applicable to the electronic passport application.
OP_005	A virtual or physical stop/go device of some kind (red/green lights, turnstile, etc.) must allow passengers not recognised in the BCA or regarded as suspect to be prevented from leaving the control area without examination by monitoring staff	Not applicable to the electronic passport application.
OP_006	Travellers moving through the BCA should be individually identified and relevant data presented on a monitor's screen next to the image of the moving passenger.	Not applicable to the electronic passport application.
OP_007	The monitoring system should link alerts to individual travellers such that a monitoring officer can understand the nature of the alert and identify the relevant traveller in the moving queue.	Not applicable to the electronic passport application.
OP_008	Travellers must be able to transmit personal biographic and biometric data (together with vehicle data, if required) either via a kiosk or a mobile device before approaching the border control area/Biometric Capture Area.	Not applicable to the electronic passport application.

OP_009	The BCA system shall parse the data coming in from eMRTD scans and apply testing algorithms such as MRZ checksum, digital hash verifications and country/document signing certificates and report against a traveller if any defects are detected	Not applicable to the electronic passport application.
OP_010	The BCA system must submit certain data to a simulated national border control databases and simulated Interpol Stolen and Lost Travel Document (SLTD) database and link the results to the traveller as they proceed through the BCA.	Not applicable to the electronic passport application.
OP_011	The passport data capture system should warn a traveller if the passport data cannot be fully captured; or if checksums in the MRZ do not match expected values; or if automatic checks suggest the document is defective; or the document or traveller is not eligible to use the BCA	Not applicable to the electronic passport application.
OP_012	The eMRTD system shall send whatever data is available, including a facial image of the traveller taken at that instant to the BCA system with an alert at the document or traveller requires manual inspection.	Not applicable to the electronic passport application.

2.1.3 Set C – Operational Requirements – Mobile Devices

Not applicable to the electronic passport application.

2.1.4 Set D – Human-Machine Interface Requirements

2.1.4.1 Enrolment kiosk

Req-ID	Description	Status
Req_ui_01	The kiosk should show command for the traveller on a touch screen.	Not applicable to the electronic passport application.
Req_ui_02	The kiosk should be able to support many languages in order to communicate efficiently with travellers: a. The system should automatically adjust the language to the user taking into account his/her nationality. b. The system should be able to provide the user with an option of language selection.	Not applicable to the electronic passport application.
Req_ui_03	Instructions on the screen should be supported with pictograms. Commands on the screen should help intuitively navigate through the software and capture biometric data needed for	Not applicable to the electronic passport application.

	enrolment. Text could be kept to a minimum.	
Req_ui_04	The kiosk should verify identity of the passenger by means of his/her travel document and face matching or fingerprint matching (not mandatory when registered in EES)	Not applicable to the electronic passport application.
Req_ui_05	Face of the passenger should be displayed on the screen during face matching process	Not applicable to the electronic passport application.
Req_ui_06	The kiosk should display a consent form to the traveller. The passenger must agree on the biometric enrolment. If not the enrolment is terminated.	Not applicable to the electronic passport application.
Req_ui_07	The kiosk should allow medium selection, where to store newly acquired biometric data.	Not applicable to the electronic passport application.

2.1.4.2 Air Border Surveillance Station

Req-ID	Description	Status
Req_ui_08	The station should be supervised by a single operator	Not applicable to the electronic passport application.
Req_ui_09	The station should display information on two screens: a) First screen should show passengers moving inside corridor with assigned temporary IDs. Each passenger should be surrounded by a rectangular. Green colour of the rectangular means correct verification, red means unverified person. Black colour means verification problem. Red and black rectangular could flash. b) Second screen should show faces of passengers. Colours of the photo frames indicate verification result (colour scheme as above).	Not applicable to the electronic passport application.
Req_ui_10	Login to the station must be possible only for an authorized operator.	Not applicable to the electronic passport application..
Req_ui_11	Language of the software should be selected from at least four languages (EN, GE, FR, PL)	Not applicable to the electronic passport application.
Req_ui_12	The station should be able to show a total verification score for each passenger.	Not applicable to the electronic passport application.
Req_ui_13	The station could be able to show matching scores for particular biometric sensors	Not applicable to the electronic passport application.
Req_ui_14	The station should be able to show the reason of verification problem	Not applicable to the electronic passport application.
Req_ui_15	The border guard should be able to set at least following parameters:	Not applicable to the electronic passport application.

	a. Show scores – Yes/No b. Show reason of verification problem – Yes/No c. Privacy filters – high/medium/low.	
Req_ui_16	The operator must be able to stop the operation of biometric corridor with a single click.	Not applicable to the electronic passport application.
Req_ui_17	The station should gather travellers statistics including entry/exit and transaction performance time.	Not applicable to the electronic passport application.
Req_ui_18	The important information (anonymised) should be stored in a log file, at least: border checks date, transaction time, total verification score, scores obtained for matching in particular biometric sensors.	Not applicable to the electronic passport application.

2.1.5 Set E – Requirements List from Stakeholder Workshops

2.1.5.1 Pre-conditions

Req-ID	Description	Status
Req_pc_01	The system should be able to check if the traveller is eligible to use the biometric.	The passport contains information on issuing country, nationality and birth date.
Req_pc_02	The system should be able to support many languages in order to communicate efficiently with travellers.	Not applicable to the electronic passport application.
Req_pc_03	The system should automatically adjust the language to the user taking into account his/her nationality	Not applicable to the electronic passport application.
Req_pc_04	The system should be able to provide the user with an option of language selection	Not applicable to the electronic passport application.
Req_pc_05	The system should be able to check EU/CH/EEA and pre-vetted TCN passengers.	The passport contains information on issuing country, nationality and birth date..
Req_pc_05a	The system should be able to perform face matching of a traveller in no more than 15 seconds	Not applicable to the electronic passport application.
Req_pc_05b	The system should be able to perform fingerprint matching of one traveller in no more than 45 seconds	Not applicable to the electronic passport application.
Req_pc_06	The system should allow for the registration of users from travellers who are 18 years old or above.	The passport contains information on issuing country, nationality and birth date.

2.1.5.2 Kiosk/enrolment

Req-ID	Description	Status
Req_enrol_01	The system should allow a passenger to use the PROTECT solution for as long as his/her travel document is valid.	fulfilled
Req_enrol_02	The enrolment kiosks should be located at arrival halls at airports	Not applicable to the electronic passport application

Req_enrol_03	The enrolment kiosks could be located at departure halls at airports	Not applicable to the electronic passport application
Req_enrol_04	The system should allow the traveller for registration to the PROTECT solution via the mobile application.	Not applicable to the electronic passport application
Req_enrol_05	The system should be able to securely capture biometric data.	Not applicable to the electronic passport application
Req_enrol_06	The system should be able to securely capture biometric data	Not applicable to the electronic passport application
Req_enrol_07	The system should be able to scan a travel document in enrolment kiosk	Not applicable to the electronic passport application
Req_enrol_08	The system should be able to verify travel documents in kiosk.	The application is ICAO compliant to LDS 1.8 and 2.0. Therefore any passport reader with certificates can check the authenticity
Req_enrol_09	The system should be able to check the document's authenticity in kiosk	The application is ICAO compliant to LDS 1.8 and 2.0. Therefore any passport reader with certificates can check the authenticity
Req_enrol_10	The system should be able to enrol gait pattern during enrolment process	Gait pattern requires a large area. This is currently simulated by walking through the gait recognition area of the BCA for enrolment.
Req_enrol_11	The border guard should be able to supervise the enrolment process	Not applicable to the electronic passport application
Req_enrol_12	The system should allow all types of travellers to enrol their biometric attributes at kiosk in order to use the PROTECT system	Not applicable to the electronic passport application

2.1.5.3 Risk management

Req-ID	Description	Status
Req_rm_01	The system should be able to show the information on consulting databases as a prompt for BG officer concerning the travellers' eligibility to cross the border.	Not applicable to the electronic passport application.
Req_rm_02	The system should be able to query simulated EES database to see where the TCN passenger travelled before.	Not applicable to the electronic passport application
Req_rm_03	The system should be able to provide information who is expected at the border control post on the basis of the data provided by a traveller via mobile application in advance of the journey	Not applicable to the electronic passport application
Req_rm_04	The system should be able to perform risk assessment after a traveller informs about his/her intentions to cross the border before entering the corridor	Not applicable to the electronic passport application
Req_rm_05	The system should be able to conduct risk profiling on the basis of simulated ETIAS and NFP data.	Not applicable to the electronic passport application
Req_rm_06	The system should be able to conduct risk profiling via an external module/system.	Not applicable to the electronic passport application

2.1.5.4 Mobile device

Req-ID	Description	Status
Req_md_01	The system should be able to store biometric data on a traveller's mobile device.	Not applicable to the electronic passport application
Req_md_02	The system should digitally sign and encrypt biometric data stored in a mobile device.	Not applicable to the electronic passport application
Req_md_03	The system should enable proper safeguarding against stolen documents.	Document number of the ePassport application can be taken in a list of stolen documents
Req_md_05	The system should be able to inform the traveller if he/she can use the PROTECT solution via the mobile application	Not applicable to the electronic passport application
Req_md_06	The system should allow TCNs to provide additional data via the mobile application (before every border crossing)	Not applicable to the electronic passport application
Req_md_07	The system should not inform a traveller about his/her biometric verification score	Not applicable to the electronic passport application

2.1.5.5 ePassport

Req-ID	Description	Status
Req_epass_01	Biometric data shall be protected against unauthorized alteration	Fulfilled, following ICAO spec protected by EAC

2.1.5.6 Biometric corridor

Req-ID	Description	Status
Req_bc_01	The biometric corridor should meet the health and safety regulations.	Not applicable to the electronic passport application
Req_bc_02	The system should be adaptable to limited space	Not applicable to the electronic passport application
Req_bc_03	The system should be able to direct unrecognized travellers to border guards	Not applicable to the electronic passport application
Req_bc_04	The system should be able to notify border guards about suspicious or unrecognized travellers.	Not applicable to the electronic passport application
Req_bc_05	The biometric corridor should be equipped with a barrier at its end.	Not applicable to the electronic passport application
Req_bc_06	The barrier at the end of the corridor should be left open if security requirements enable it.	Not applicable to the electronic passport application
Req_bc_07	The system could be able to detect people inside the biometric corridor which are consciously not compliant with the instructions	Not applicable to the electronic passport application
Req_bc_08	The corridor's interior should be equipped with useful passenger information	Not applicable to the electronic passport application

Req_bc_09	The system should give border guard the possibility to interrupt the process at any time.	Not applicable to the electronic passport application
Req_bc_10	The system should be able to track people moving in the corridor	Not applicable to the electronic passport application

2.1.5.7 Crowd Control

Req-ID	Description	Status
Req_crowd_01	The system should be able to notify a traveller when he/she can enter the corridor	Not applicable to the electronic passport application
Req_crowd_02	The system should be able to notify a traveller when his/her biometrics are collected	Not applicable to the electronic passport application
Req_crowd_03	The system should display a consent form to a traveller.	Not applicable to the electronic passport application
Req_crowd_04	System should ensure 1:1 matching between data stored on mobile device with biometrics collected in the corridor.	Not applicable to the electronic passport application

2.1.5.8 Human-Machine Interface (HMI)

Req-ID	Description	Status
Req_hmi_02	The system should be able to show a verification score of passengers on border guard's interface.	Not applicable to the electronic passport application
Req_hmi_03	The system should be able to show green light on BG interface to indicate that a traveller is clear to go	Not applicable to the electronic passport application
Req_hmi_04	The system should be able to show red light on BG interface to indicate that a traveller must be stopped.	Not applicable to the electronic passport application
Req_hmi_05	The system should be able to provide statistics on transaction times.	Not applicable to the electronic passport application
Req_hmi_06	The system should be able to display faces of people going through the corridor on BG interface.	Not applicable to the electronic passport application
Req_hmi_07	The system should be able to display information on unverified cases on BG interface.	Not applicable to the electronic passport application

2.1.6 General Stakeholder Feedback

Issue-ID	Description	Status
Issue_01	The consortium should address the problem of possible system failure and its impact (preparation of contingency plans and procedures in case of such event)	Not applicable to the electronic passport application
Issue_02	Project partners should consider, at the system design stage, the differences between the border types and adjust the system to specific requirements.	Not applicable to the electronic passport application

Issue_03	The quality of the documents used by TCNs happens to be poor, which prolongs the verification process.	The epassport app will be of high quality
Issue_04	There could be problems in 'on the move' verification if the distance between the sensor and a traveller is excessive.	Not applicable to the electronic passport application
Issue_05	Lack of interoperability between customs and border guard databases	Not applicable to the electronic passport application
Issue_06	The verification of the people inside a vehicle can be problematic (if the vehicle is checked via ABC)	Not applicable to the electronic passport application
Issue_07	Legal regulation of data storage has to be taken into account on the national and EU level.	Data storage is done with the same security measures as with normal passports
Issue_08	Some travellers could have problems with providing required data if they will not get any help from assistance personnel. Problematic might be also using such a novel solution as biometric corridor.	Not applicable to the electronic passport application
Issue_09	There is a need for a legislation framework that will fully regulate the use of multiple biometric features	Not applicable to the electronic passport application

2.2 Life cycle phases

The epassport app will support the following life cycle phases:

2.2.1 Initialization phase

This is the phase, where all the components e.g. file containers, key objects or password objects are create.

2.2.2 Personalization phase

Within personalization phase the containers for files, keys or passwords are filled with cardholder's data.

2.2.3 Usage phase

During this phase the cardholder uses the product for travelling.

2.3 LDS Requirements

There should be applications for the following purposes

- MF Application
- LDS logical data system
- Travel Records Application
- VISA Records Application
- Additional Biometric Application

All requirements from [11] have to be fulfilled.

3 Physical Device

3.1 Secure Crypto Controller

- Chip: Infineon SLE78CLFX6280P
- Security Certification: Common Criteria EAL 6+
- Free Flash Memory: 300KB

3.2 Javacard Operating System

- Veridos SmartCafe Expert 7.1 I
- Java Card Spec 3.0.4
- Global Platform 2.2 and Amendment A,D,E
- Security Certification: Common Criteria EAL 5+

4 Data structures

For a detailed description of the objects and data structures used in the certificates the reader is referred to [8] und [9]. These specifications by BSI explain the process of using EAC certificates and the purpose of each object.

4.1 Card Verifiable Certificates used by EACv1

Card verifiable certificates are used to authorize the reading device to the security chip. The certificates have to be presented to the chip one by one and are verified by 2 public keys stored on the chip. The Card Holder Authorization Template (CHAT) and Object Identifier (OID) describe the actual access right of the reading device.

4.1.1 DV Certificate

The DV Certificate is the first one to be loaded into the card during the EAC process. It contains the public key of the IFD certificate.

Tag	Length	Value			
7F4E'	L7F4E	Structure of certificate			
		Tag	length	value	
	5F29'	01'	00'		
	42'	06' - '10'	card authorite reference		
	7F49'	L7F49	Public key object		
			Tag	length	value
			06'	0A'	Object identifier id_TA_ECDSA_(SHAn)
			86'	L86	'04' x(PuK.DV) y(PuK.DV)
	5F20'	L 5F20	Card holder reference CHR		
	7F4C'	L 7F4C	CHAT		
			Tag	length	Value
			06'	09'	id_IS
53'			01'	access bitmap	

	5F25	06'	CED							
	5F24	06'	CXD							
	65	L 65								
			Tag	length	value					
			73	L 73						
				tag	length	Value				
				06'	09	Object identifier for travel records				
				53'	01	access bitmap				
			73	L 73						
				tag	length	Value				
				06'	09	Object identifier for VISA records				
				53'	01	access bitmap				
			73	L 73						
				tag	length	Value				
				06'	09	Object identifier for additional biometrics				
				53'	01	access bitmap				
			5F37	L 5F37	ECDSA signature					

Figure 1 DV certificate that imports an transient ECDSA key usable for verify certificate.

4.1.2 IFD Certificate

The IFD certificate contains the public key of the epassport reader.

Tag	Length	Value			
7F4E'	L7F4E	Structure of certificate			
		Tag	length	value	
		5F29'	01'	00'	
		42'	06' - '10'	card authorite reference	
		7F49'	L7F49	Public key object	
			Tag	length	value
			06'	0A'	Object identifier id_TA_ECDSA_(SHAn)
			86'	L86	04' x(PuK.IFD) y(PuK.IFD)
		5F20'	L 5F20	Card holder reference CHR	
		7F4C'	L 7F4C	CHAT	
			Tag	length	Value
			06'	09'	id_IS
			53'	01'	access bitmap
		5F25	06'	CED	
		5F24	06'	CXD	
		65	L 65		
			Tag	length	value

		73	L 73			
			tag	length	Value	
			06'	09	Object identifier for travel records	
			53'	01	access bitmap	
		73	L 73			
			tag	length	Value	
			06'	09	Object identifier for VISA records	
			53'	01	access bitmap	
		73	L 73			
			tag	length	Value	
			06'	09	Object identifier for additional biometrics	
			53'	01	access bitmap	
5F37	L 5F37	ECDSA signature				

Figure 2 IFD certificate that imports a transient ECDSA key usable for external authenticator.

4.1.3 CVCA Certificate

The CVCA certificate is used to change the root keys on the chip.

Tag	Length	Value				
7F4E'	L7F4E	Structure of certificate				
		Tag	length	value		
		5F29'	01'	00'		
		42'	06' - '10'	card authority reference		
		7F49'	L7F49	Public key object		
				Tag	length	value
				06'	0A'	Object identifier id_TA_ECDSA_(SHAn)
				81'	L 81	prime P
				82'	L 82	coefficient A
				83'	L 83	coefficient B
				84'	L 84	04' generator G
				85'	L 85	order R
				86'	L 86	'04' x(PuK.DV) y(PuK.DV)
		5F20'	L 5F20	Card holder reference CHR		
		7F4C'	L 7F4C	CHAT		
				Tag	length	Value
				06'	09'	id_IS
				53'	01'	access bitmap
		5F25	06'	CED		
		5F24	06'	CXD		
		65	L 65			

		Tag	length	value		
		73	L 73			
				tag	length	Value
				06'	09	Object identifier for travel records
				53'	01	access bitmap
		73	L 73			
				tag	length	Value
				06'	09	Object identifier for VISA records
				53'	01	access bitmap
		73	L 73			
				tag	length	Value
				06'	09	Object identifier for additional biometrics
				53'	01	access bitmap
		5F37	L 5F37	ECDSA signature		

Figure 3 Link certificate that links to an ECDSA root key.

5 LDS2.0 Application Interface

The LDS2 specification [10] foresees the following file structure on the security chip:

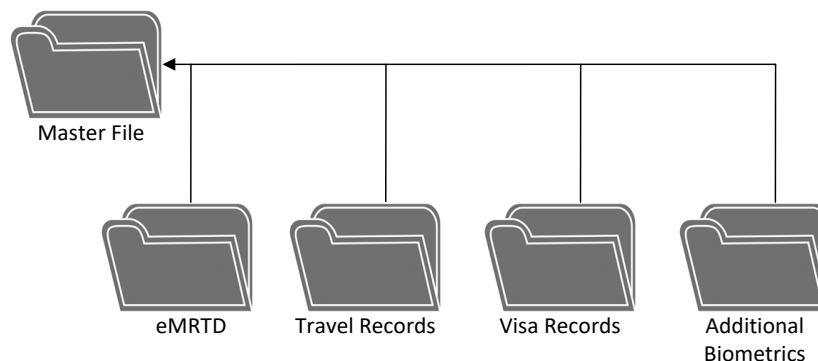


Figure 4 File structure of Logical Data Structure 2

To support the requirements from [10], 3 separate javacard applications have to be created. These applications share the security status established in the master file application.

5.1 The Master File

5.1.1 Registry of Children

A child in this context is a dedicated file or application according to [10]. Child can be the 3 applications defined in [11].

Registry of children requires the class to implement either the interface “IAppProcess” or “IMasterAppProcess”. These interfaces contain the methods for selection, process and deselection of application. The interface “IMasterAppProcess” contains an additional method for selection of the parent application.

5.1.1.1 List of Children

The master file contains a linked list with objects that contains entries with the data of registered children. These entries consist of the child's AID and an object that links to the child's process routine.

5.1.1.2 Actual selected Children

Furthermore the master file contains a transient object that stores the actual selected children or a link to the master-file's process routine.

5.1.2 Register Children

The child applications are registered during the personalization phase.

Whenever the host transfers the AID of a child application to the master application, the master application tries to register the children.

First the master application searches within the registry for an existing child which contains the same AID. In case no entry is found, the master tries to retrieve the shareable interface for this AID. On successful retrieval of the shareable object, the master application creates a new node within the registry for this child.

5.1.3 Selection of a Children

On issuing a select by AID command to the master application, the application tries to find the entry with the given AID. When found, the object of the shareable process routine is verified to be not null.

5.1.4 File components

The master application contains following file containers:

- EF.DIR (optional)
- EF.CardAccess
- EF.CardSecurity

Details on file ids and file types are specified in [8] and [11].

5.1.5 Key components

The master file is used to execute the privacy protocol and the EAC protocol in version 2. To drive this protocol, different keys are necessary.

5.1.5.1 Nonvolatile Key Components

Following nonvolatile keys are used by these protocols.

- Symmetric PACE passwords or keys
- Private elliptic curve key for Chip Authentication
- Public elliptic curve key for Terminal Authentication

5.1.5.2 Transient Key Components

Following transient keys are used by these protocols.

- Ephemeral key pair for PACE map nonce
- Ephemeral key pair for PACE mutual authentication
- Symmetric Keys (3DES or AES) for secure messaging

5.1.6 Protocol Handlers

The master application supports protocol handlers for following protocols:

- BAC
- PACE
- EAC Version 1

Additionally there are methods for setup of secure messaging by the child applications.

5.1.6.1 Interfaces for Protocol Handlers

The shareable interface for the protocol handlers makes the authentication commands accessible for the child applications. This interface contains command interpreter functions for following commands:

- External Authenticate
- Get Challenge
- General Authenticate
- Verify Certificate

The command structure of these commands is specified in [10] and [8].

The parameter passed to these functions is the current received APDU. All functions return the length of the data to send as response.

5.1.7 Command Handlers

The master application supports following command handlers

General Authenticate (PACE, EAC)

- External Authenticate
- General Authenticate
- MSE:SET AT
- MSE:SET DST
- Read Binary
- Select File
- Select Application
- Verify Certificate

The command structure of these commands is specified in [10] and [8].

5.2 The ICAO Application

The ICAO application can be used either as an EACv1 passport or an LDS-2.0 passport application. Therefore the application can be selected either with plain text APDU or through an established secure messaging channel.

5.2.1 Selection of a Children

Either the select parent command or the select MF command is redirected to the Master File application. Additionally all select commands with unknown AID are redirected to the Master File application.

5.2.2 File components

The ICAO application usually contains all file containers defined by the ICAO 9303 specification. These are:

- EF.CVCA
- EF.DG1 ... EF.DG20
- EF.SOD

Details on file ids and file types are specified in [8].

5.2.3 Protocol Handlers

After selection of the ICAO application, a secure channel can be established using BAC and EACv1 protocol. The protocol implementation is resided within the MF-application.

The ICAO application supports the ACTIVE AUTHENTICATE command with either RSA or ECDSA keys.

5.2.4 Command Handlers

The ICAO application supports the following command handlers

- Select File
- Select Application
- Get Challenge
- Mutual Authenticate BAC
- MSE:SET AT
- MSE:SET KAT
- Verify Certificate (protocol within MF)
- External Authenticate EACv1 (protocol within MF)
- General Authenticate EACv1 (protocol within MF)
- Read Binary
- Active Authenticate

The command structure of these commands is specified in [10] and [8].

5.3 The Travel Record Application

The travel record application can be used either as instance for the travel records or as instance for VISA records.

5.3.1 Selection of a Children

Either the select parent command or the select MF command is redirected to the Master File application. Additionally all select commands with unknown AID are redirected to the Master File application.

5.3.2 File components

5.3.2.1 Components for usage as travel record application

The Travel Record application contains following file containers:

- EF.Certificates
- EF.EntryRecords

- EF.ExitRecords

Details on file ids and file types are specified in [11].

5.3.2.2 Components for usage as VISA application

The VISA application contains following file containers:

- EF.Certificates
- EF.VisaRecords

Details on file ids and file types are specified in [11].

5.3.3 Protocol Handlers

The authentication and establish of secure channel using PACE and EAC is done within the MF application.

5.3.4 Command Handlers

The Travel Records application supports following command handlers

- Select File
- Select Application
- Append Record
- Search Record
- Read Record

The command structure of these commands is specified in [10] and [8].

5.4 The Additional Biometric Application

The additional biometric application is used to store additional biometrics.

5.4.1 Selection of a Children

Either the select parent command or the select MF command is redirected to the Master File application. Additionally all select commands with unknown AID are redirected to the Master File application.

5.4.2 File components

The additional biometric application contains following file containers:

- EF.Certificates
- EF.Biometrics 1 .. EF.Biometrics n

Details on file ids and file types are specified in [11].

5.4.3 Protocol Handlers

The authentication and establish of secure channel using PACE and EAC is done within the MF application.

5.4.4 Command Handlers

The Travel Records application supports following command handlers

- Select File
- Select Application
- Append Record

- Search Record
- Read Record
- Read Binary
- Write Binary (write once)

The command structure of these commands is specified in [10] and [8].

6 Initialization and Personalization

Neither the LDS1 [7], BSI[8]/[9] or LDS2 specification [11] does specify the initialization and personalization phase of a passport applet. This is left to the manufacturer. This section will describe the initialization and personalization of the PROTECT ePassport app.

6.1 Installation of the Applets

Several applet instances are required for a complete LDS2.0 passport.

These applet instances are:

- The master file application MF
- The ICAO application
- One instance of the travel record application for TR
- Another instance of the travel record application for VISA
- The additional biometrics application.

The MF application must be created as the first instance.

Each instance of the LDS2.0 applets is installed using the global platform command INSTALL FOR INSTALL.

6.2 Initialization of the applets

The general process of installing javacard applets on a global platform card is described in the global platform card specification [12].

In order to serve as a LDS2.0 passport, these needed resources must be created.

Each application must be initialized separately. In addition, initialization should be completed in each application using the personalize-end command.

Applications to be initialized are:

- The MF Application instance
- the ICAO Application instance
- the TR Application instance
- the VISA Application instance
- the Additional Biometrics Application instance

6.2.1 Initialization of the MF Application instance

6.2.1.1 Components to be created.

In the MF application, both cryptographic components and file containers must be created.

The file containers are as following:

- EF.ATR,
- EF.DIR,
- EF.CardAccess,
- EF.Cardsecurity

The cryptographic components are as following:

- Elliptic curve used by PACE
- PACE passwords
- Elliptic curve used by CHIP Authenticate
- Private key for CHIP Authenticate
- The certificate containing the public key for TERMINAL Authenticate
- The current date for TERMINAL Authenticate

6.2.1.2 Finalize the initialization

Command APDU for finalize initialization: 00 40 04 00 02 4f 00

After sending this command to the selected applet, no more resources can be created.

6.2.2 Initialization of the ICAO application instance

6.2.2.1 Components to be Created.

Both file containers and cryptographic components must be created in the ICAO application.

The file containers are as following:

- EF.DG1 to EF.DG20
- EF.SOD
- EF.CVCA

The cryptographic components are as following:

- The BAC key set for mutual authenticate
- Elliptic curve used by ACTIVE AUTHENTICATE
- The private key for ACTIVE AUTHENTICATE

or

- The private RSA key for ACTIVE AUTHENTICATE

6.2.2.2 Finalize the initialization

Command APDU for finalize initialization: 00 40 04 00 02 4f 00

After sending this command to the selected applet, no more resources can be created.

6.2.3 Initialization of the Travel Record application instance

6.2.3.1 Components to be created

There are only file containers must be created in the Travel Record application.

The file containers are as following:

- EF.EntryRecords
- EF.ExitRecords
- EF.Certificate

6.2.4 Initialization of the VISA Record application instance

The initialization of the VISA application is similar to the initialization of the travel record application.

6.2.5 Initialization of the Additional Biometrics application instance

6.2.5.1 Components to be Created.

There are only file containers must be created in the Travel Record application.

The file containers are as following:

- EF.Certificate
- EF.Biometrics 1 .. EF.Biometrics n

6.3 Personalization of the Applets

In order to serve as a LDS2.0 passport, both the publisher's data and the cardholder's data must be personalized. Each application must be personalized separately. In addition, personalization must be completed in each application using Personalize-End.

Applications to be personalized are:

- The MF Application instance
- the ICAO Application instance
- the TR Application instance
- the VISA Application instance
- the Additional Biometrics Application instance

6.3.1 Personalization of the MF Application instance

6.3.1.1 Components to be personalized.

In the MF application, both cryptographic components and file containers must be personalized.

The file containers are as following:

- EF.ATR,
- EF.DIR,
- EF.CardAccess,
- EF.Cardsecurity

The cryptographic components are as following:

- Elliptic curve used by PACE
- PACE passwords
- Elliptic curve used by CHIP Authenticate
- Private key for CHIP Authenticate
- The certificate containing the public key for TERMINAL Authenticate

- The current date for TERMINAL Authenticate

6.3.2 Personalization of the ICAO application instance

6.3.2.1 Components to be personalized.

In the ICAO application, both cryptographic components and file containers must be personalized.

The file containers are as following:

- EF.DG1 to EF.DG20
- EF.SOD
- EF.CVCA

The cryptographic components are as following:

- The BAC key set for mutual authenticate
- Elliptic curve used by ACTIVE AUTHENTICATE
- The private key for ACTIVE AUTHENTICATE

or

- The private RSA key for ACTIVE AUTHENTICATE

6.3.3 Personalization of the TR Application instance

6.3.3.1 Components to be personalized.

Usually, there are no components to be personalized. So the only task is to end personalization and switch to usage state. However the append record command can also be used during personalization to write certificates to the application.

6.3.4 Personalization of the VISA Application instance

6.3.4.1 Components to be personalized.

Usually, there are no components to be personalized. So the only task is to end personalization and switch to usage state. However the append record command can also be used during personalization to write certificates to the application.

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Security Sensitivity Assessment

Pervasive and UseR Focused BiomeTrics BordEr ProjeCT (PROTECT) H2020 – 700259

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TECHNICAL REPORT

Logical Data Structure (LDS) for Storage of Data in the Contactless IC

Doc 9303-10 LDS 2 – New Applications

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Technical Report

Doc 9303-10 LDS2 New Applications V20.7

Date : April 3, 2018

Release Control

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12.8	20 Sept 15	Added AIDs, incorporated into TR format
12.9	09 Oct 15	Visa records simple TLV tags changed to BER TLV, multiple editorial changes
14.0	09 Nov 15	
15.0	25 July 16	Incorporate agreed changes from April 2016 Wellington meeting
16.0	25 Oct 16	Incorporate agreed changes from October 2016 Berlin meeting
16.1		From Berlin meeting added Proprietary search criteria (SearchRecord-Proposal-GD-07-12-2015.pdf). removed annex D example Option 1 SEARCH RECORD
17.0	1 April 2017	Incorporate changes from WG3 March 2017 Milpitas Meeting Add initial Biometrics Application sections
18.0	11 Aug 2017	Incorporate changes from 8 April 2017 telco, NTWG meeting 26 April, and additional posted comments
19.0	11 Jan 2018	Incorporate changes from Oct 2017 Paris meeting Added proprietary WRITE BINARY section 6.4.1 Added SEARCH RECORD method according to ISO/IEC 7816-4:2013/DAmD 1
20.6	14 Mar 2018	Incorporate changes from March 2018 Tokyo meeting. Additional Biometrics transparent files handling part updated. Added proprietary File and Memory Management command Proprietary SEARCH RECORD command option removed
20.7	03 April 2018	Second round reviews from Tokyo meeting. Finalized EF.Biometrics and introduction of proprietary FMM command for memory management

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1. INTRODUCTION

This document specifies the file structures required to support the ICAO LDS2 project consisting of three additional and optional applications:

- Travel records (stamps);
- Visa records; and
- Additional biometrics.

The eMRTD may support one, several or all of these applications.

1.1 Scope and Purpose

The Doc 9303-10 LDS2 describes the file structures and application framework aspects of LDS2 and is intended to be integrated into a subsequent Technical Report on LDS2 along with other material covering the PKI and access protocols for the additional applications.

1.2 Assumptions

It has been assumed that the reader is familiar with ICAO Doc 9303 seventh edition and that this document is used in conjunction with the companion LDS2 documents;

- TR LDS2 – Protocols v0.8, April 2017;
- TR LDS2 – PKI Draft 0.8 October 2016.

1.3 Terminology

This document uses the terminology from Doc 9303 7th edition.

1.4 Data Elements Encoding Rules

The context of eMRTD uses two different tag allocation schemes for application class tag, such as defined in Doc9303-10 (LDS tag) and ISO/IEC 7816-6 (Interindustry tag).

- EF.ATR/INFO and EF.DIR use interindustry tag allocation scheme
- DFs with their containing EFs use LDS tag allocation scheme.

Interindustry tags specified in this document are used in LDS context, so coexistent tag allocation scheme is not required.

2. ELEMENTARY FILES

2.1 EF.ATR/INFO (MANDATORY)

EF.ATR/INFO must be located in the MF. The short EF identifier at MF level is '01'.

Table 1: EF.ATR/INFO

File Name	EF.ATR/INFO
File ID	'2F01'
Short EF Identifier	'01'
Select / FMM Access	ALWAYS
Read Access	ALWAYS
Write/Update/Erase Access	NEVER
File structure	Transparent
Size	Variable

The contents of the EF.ATR/INFO can be retrieved by using a SELECT command followed by READ BINARY command. The READ BINARY command response data field contains the content of the EF.ATR/INFO.

Table 2: Data elements of EF.ATR/INFO

Tag	Length	Value	Notes
'47'	'03'	Card capabilities	
		byte 1 - first software function	b8=1: DF selection by full DF name b7 to b4 and b1 are out of scope of Doc 9303 b3=1: short EF identifier supported b2=1: record number supported
		byte 2 - second software function	b8 and b5 are out of scope of Doc 9303 b7 to b6=00: one-time write b4 to b1=0001: one byte data unit size
		byte 3 - third software function	b8=1: command chaining supported b7=1: Extended Lc and Le fields supported b6=1: Extended length information in EF.ATR/INFO b5 to b1 are out of scope of Doc 9303
'7F66'	'xx'	Extended length information	
		Tag	Length
		Value	Notes
		'02'	'02'
		Positive integer - the maximum number of bytes in the command data field	MUST be at least 1000
		'02'	'02' or '03'
		Positive integer - the maximum number of bytes expected in the response APDU	MUST be at least 1000

Note: Further data objects MAY be present in EF.ATR/INFO.

Note: EF.ATR/INFO uses standard tag allocation scheme as defined in ISO/IEC 7816-4

2.2 EF.CardSecurity (CONDITIONAL)

EF.CardSecurity is a transparent EF contained in the MF. The short EF identifier at MF level is '1D'.

Table 3: EF.CardSecurity

File Name	EF.CardSecurity
File ID	'011D'

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Short EF Identifier	'1D'
Select Access	PACE
Read Access	PACE
Write/Update/Erase Access	NEVER
File structure	Transparent
Size	Variable

EF.CardSecurity contained in the MF is REQUIRED if

- PACE with Chip Authentication Mapping is supported by the IC;
- Terminal Authentication in the MF is supported by the IC; or
- Chip Authentication in the MF is supported by the IC.

and MUST contain

- ChipAuthenticationInfo as required by Chip Authentication;
- ChipAuthenticationPublicKeyInfo as required by PACE-CAM/Chip Authentication;
- TerminalAuthenticationInfo as required by Terminal Authentication;
- the SecurityInfos contained in EF.CardAccess.

2.3 EF.DIR (REQUIRED)

EF.DIR MUST be located in the MF. The short EF identifier at MF level is '1E'.

Table 4: EF.DIR

File Name	EF.DIR
File ID	'2F00'
Short EF Identifier	'1E'
Select Access	ALWAYS
Read Access	ALWAYS
Write/Update/Erase Access	NEVER
File structure	Transparent
Size	Variable

EF.DIR MUST indicate a list of applications supported by the eMRTD. It MUST contain a set of application templates and / or application identifier Dos in any order.

Table 5: EF.DIR Format

Tag	L	Value			Description
'61'	'09'				eMRTD Application Template
		Tag	L	Value	eMRTD Application 'International AID': 'A0 00 00 02 47 10 01'
		'4F'	'07'	'A0 00 00 02 47 10 01'	
'61'	'09'				Travel Records 'Application Template'
		Tag	L	Value	Travel Records 'International AID': 'A0 00 00 02 47 20 01'
		'4F'	'07'	'A0 00 00 02 47 20 01'	
Tag	L	Value			
'61'	'09'				Visa Records 'Application Template'
		Tag	L	Value	Visa Records 'International AID': 'A0 00 00 02 47 20 02'
		'4F'	'07'	'A0 00 00 02 47 20 02'	
Tag	L	Value			
'61'	'09'				Additional Biometrics 'Application Template'
		Tag	L	Value	Additional Biometrics 'International AID': 'A0 00 00 02 47 20 03'
		'4F'	'07'	'A0 00 00 02 47 20 03'	

Note: EF.DIR uses standard tag allocation scheme as defined in ISO/IEC 7816-4

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The Application present on the eMRTDs MUST be referenced in the EF.DIR.

2.4 EF.CardAccess (CONDITIONAL)

EF.CardAccess is a transparent EF contained in the MF. The short EF identifier at MF level is '1C'.

Table 6: EF.CardSecurity

File Name	EF.CardAccess
File ID	'011C'
Short EF Identifier	'1C'
Select Access	ALWAYS
Read Access	ALWAYS
Write/Update/Erase Access	NEVER
File structure	Transparent
Size	Variable

EF.CardAccess contained in the MF is REQUIRED if PACE is supported by the chip and MUST contain the following SecurityInfos for the PACE protocol:

- PACEInfo
- PACEDomainParameterInfo

3. TRAVEL RECORDS APPLICATION (CONDITIONAL)

The Travel Records application MAY be implemented by an issuing State or organization. The following is conditionally REQUIRED if the optional Travel Records application has been invoked.

3.1 File Structure Summary

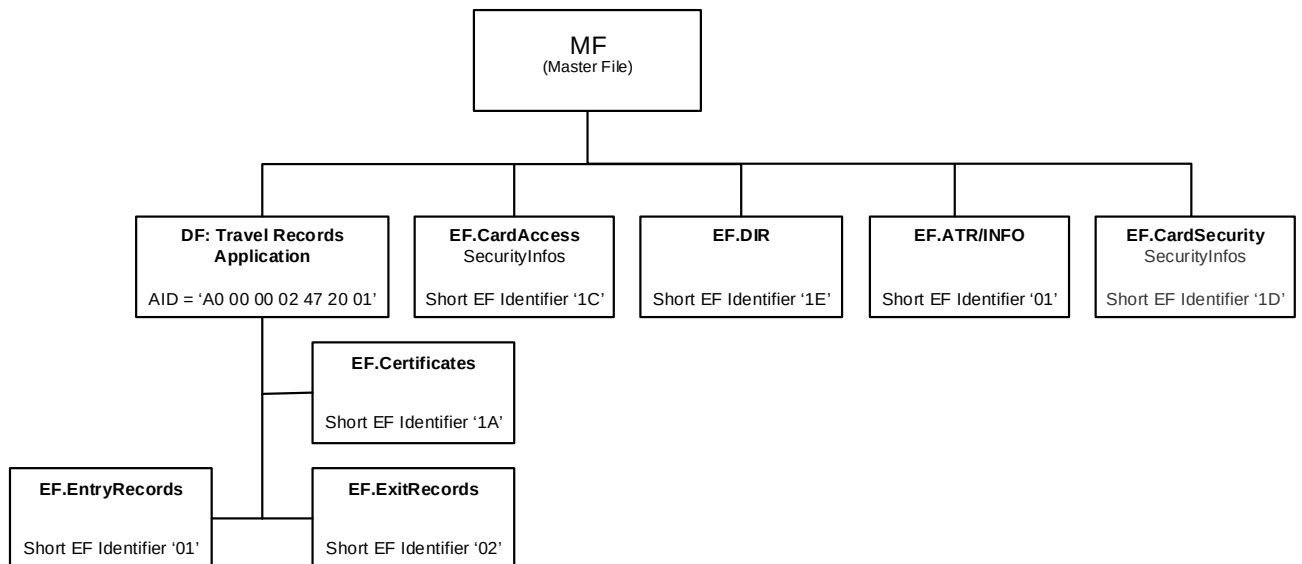


Figure 1: Travel Records Structure

3.2 EF.Certificates (REQUIRED)

The Travel Records Signer certificates are stored in an EF inside the application DF and having Linear Structure with Records of Variable Size. These certificates are intended to be used by the IS to further offline validation of the digital signatures for each record in both the EF.ExitRecords and EF.EntryRecords files.

Table 7: EF.Certificates

File Name	EF.Certificates
File ID	'011A'
Short EF Identifier	'1A'
Select / FMM Access	PACE+TA (Travel record authorization bit b3 according to table 21)
Read record / Search Record Access	PACE+TA (Travel record authorization bit b3 according to table 21)
Append Record Access	PACE+TA (Travel record authorization bit b4 according to table 21)
Write / Update Record Access	NEVER
Erase Record Access	NEVER
File structure	Linear structure with records of variable size
Size	Variable

Certificate record contains a single LDS2-TS Signer X.509 certificate data object. A Certificate record MAY be referenced by one or more Entry or Exit Travel Records.

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Table 8: EF.Certificates Record Format

Tag	Content	Mandatory /Optional	Format	Example
5F3A	Certificate serial number	M	F(22)B	'5F3A' '16' {Country code SerialNumber }
72	X.509 certificate	M	V (1000) B	'72' Len { X.509 Certificate }

Note: Interindustry tags specified in this table are used in LDS context, so coexistent tag allocation scheme is not required.

DO'5F3A MUST contain a 2 letter country code according to ISO-3166-1 (same encoding and value as X.509 certificate's subject's countryName) followed by 20 bytes certificate serial number.

Each X.509 certificate contains a set of ASN.1 encoded data elements illustrated in the table below. Detailed requirements for the X.509 Certificate can be found in Doc 9303-12 Certificate Profile Specification.

Table 9: X.509 Certificate structure example

Field	Description	Example value
Certificate		
version	Must be ver.3	2
serialNumber	unique positive integer	20 bytes max
signature	Signature algorithm	ecdsa-with-SHA256
issuer		
countryName	Issuing country name	"US"
commonName	Issuer name (9 chars max)	"DHSCA0001"
validity		
notBefore	Cert. effective date	"131225000000Z"
notAfter	Cert. expiration date	"230824235959Z"
subject		
countryName	IS country name	"US"
commonName	IS name (9 chars max)	"SFO000001"
subjectPublicKeyInfo		
Public Key Algorithm	ecPublicKey	
Subject Public Key	IS public key	ECC256 Public Key
extensions		
AuthorityKeyIdentifier		
ExtKeyUsage		
Signature Algorithm	ecdsa-with-SHA256	
Signature	Issuer's Signature	ECDSA256 signature

Note: This table is an example for illustration only. Certificate records are written to EF.Certificates located under the Travel Records application DF using the APPEND RECORD command. Certificate records can be read from EF.Certificates using READ RECORD command. Certificate records MUST NOT be updated or erased. The maximum number of records in EF.Certificates under the Travel Records application DF MUST be 254.

3.3 Application Selection

The Travel Records application MUST be selected by use of the Application Identifier (AID) as a reserved DF name. The AID MUST consist of the Registered Application Identifier assigned by ISO according to [ISO/IEC 7816-5] followed by the Proprietary Application Identifier Extension (PIX) of the Travel Records application:

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- The Registered Application Identifier is 'A0 00 00 02 47'
- The Travel Records application MUST use PIX = '20 01';
- The full AID of the Travel Records application MUST be A0 00 00 02 47 20 01.

The IC MUST reject the selection of an application if the extension for this application is absent.

3.4 Entry and Exit Records

Entry and Exit Travel Records are stored in two separate Elementary Files EF.EntryRecords and EF.ExitRecords under the Travel Records application DF with both having Linear Structure with Records of Variable Size as per [ISO/IEC 7816-4].

3.5 EF.ExitRecords (REQUIRED)

It is REQUIRED to APPEND Exit Records upon embarkation at the IS.

Table 10: EF.ExitRecords

File Name	EF.ExitRecords
File ID	'0102'
Short EF Identifier	'02'
Select / FMM Access	PACE+TA (Travel record authorization bit b1 according to table 21)
Read Record / Search Record Access	PACE+TA (Travel record authorization bit b1 according to table 21)
Append Record Access	PACE+TA (Travel record authorization bit b2 according to table 21)
Write / Update Record Access	NEVER
Erase Record Access	NEVER
File structure	Linear structure with records of variable size
Size	Variable

The content of an Exit Record is shown in the table below.

Note: Interindustry tags specified in the table below are used in LDS context, so coexistent tag allocation scheme is not required.

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Table 11: Entry / Exit Record Format

Tag	Tag	Content	Mandatory /OPTIONAL	Format	Example
5F44		Embarkation/Debarcation State (copy for SEARCH RECORD)	M	F (3) A,N,S	USA
73	Entry / Exit Travel Record (signed info)				
	5F44	Embarkation/Debarcation State	M	F (3) A	USA
	5F4C	Visa approvals, refusals, and revocations	O	V (50) A,N,S	Free-form text
	5F45	Travel date (Date of entry/exit)	M	F (8) N	20120814 (yyyymmdd)
	5F4B	Inspection authority	M	V (10) A,N,S	CBP
	5F46	Inspection location (Port of Entry/Exit)	M	V (10) A,N,S	SFO
	5F4A	Inspector reference	M	V (20) A,N,S	SFO00001234
	5F4D	Result of inspection	O	V (50) A,N,S	Free-form text
	5F49	Mode of travel	O	F (1) A	A (Air), S (Sea), L (Land)
	5F48	Duration of stay (days)	O	V (2) B	'00FF' (255 days)
	5F4E	Conditions holder is required to observe whilst in issuing State	O	V(50) A,N,S	Free-form text
5F37	Authenticity token (Signature)		M	V (140) B	'5F' '37' Len {Signature}
5F38	Reference (record number) to LDS2-TS Signer certificate in Certificates Store		M	F (1) B	'01' ...'FE'

Note 1: A = Alpha character [a..z, A..Z], N = Numeric character [0..9], S = Special character ['<' ' '], B = 8-bit Binary data (any other than A, N or S), F = fixed-length field, V = variable-length field.

The order of the data objects in a record is fixed. The IS MUST build up the record content using the data objects in the order specified in the table.

Each Record MUST contain a digital signature (Authenticity Token) calculated over the DO'73, including Tag 73 and Length. Signature is generated by the LDS2-TS Signer.

LDS2-TS Signer certificates required to verify Travel Record's signature MUST be stored in the EF.Certificates under the Travel Records application DF if not already available in the same file.

Note 2: Since LDS2-TS Signer certificates are likely to be the same in multiple Travel Records (ex., when entering and exiting a country through the same airport having only one LDS2-TS Signer), before writing/appending a new certificate to the EF.Certificates, the IS should look up the EF.Certificates for a copy of the same certificate, and reference the existing one. This will reduce the size of EF.Certificates and enable faster lookups.

Travel Records are written (appended) to EF using APPEND RECORD. Travel Records MUST NOT be altered (updated) or deleted. The maximum number of records in each EF allowed MUST be 254.

Note 3: The eMRTD does not enforce that an IS writes Entry Records only to the EF.EntryRecords, but not to the EF.ExitRecords, and vice versa.

Note 4: Embarkation/Debarcation State 3-letter code according to ISO3166-1

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3.6 EF.EntryRecords (REQUIRED)

It is REQUIRED to use Entry records during debarkation at the IS.

Table 12: EF.EntryRecords

File Name	EF.EntryRecords
File ID	'0101'
Short EF Identifier	'01'
Select / FMM Access	PACE+TA (Travel record authorization bit b1 according to table 21)
Read Record / Search Record Access	PACE+TA (Travel record authorization bit b1 according to table 21)
Append Record Access	PACE+TA (Travel record authorization bit b2 according to table 21)
Write / Update Record Access	NEVER
Erase Record Access	NEVER
File structure	Linear structure with records of variable size
Size	Variable

The structure of the entry record is identical to the structure of the exit record specified in clause 3.5.

4. VISA RECORDS APPLICATION (CONDITIONAL)

The Visa Records application MAY be implemented by an issuing State or organization. The following is conditionally REQUIRED if the optional Visa Records application has been invoked.

4.1 File Structure Summary

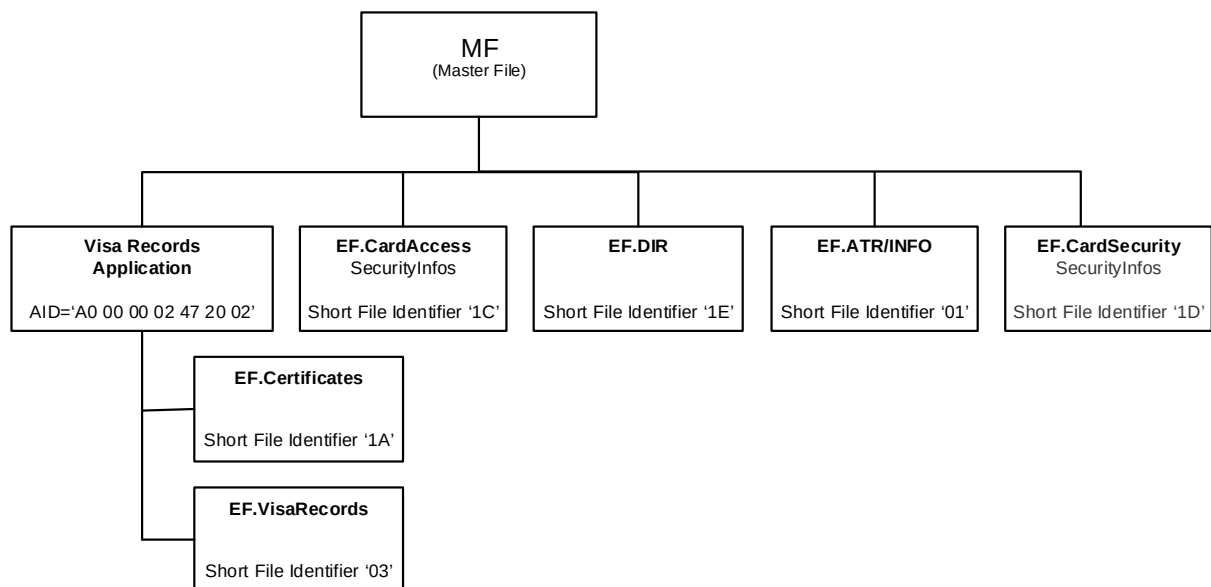


Figure 2: Visa Records Structure

4.2 EF.Certificates (REQUIRED)

The Visa Records Signer certificates are stored in EF.Certificates inside the application DF and having linear structure with records of variable size. These certificates are intended to be used by the IS to further offline validation of the digital signature for each record in the EF.VisaRecords.

Table 13: EF.Certificates

File Name	EF.Certificates
File ID	'011A'
Short EF Identifier	'1A'
Select / FMM Access	PACE+TA (Visa record authorization bit b3 according to table 22)
Read Record / Search Record Access	PACE+TA (Visa record authorization bit b3 according to table 22)
Append Record Access	PACE+TA (Visa record authorization bit b4 according to table 22)
Write / Update Record Access	NEVER
Erase Record Access	NEVER
File structure	Linear structure with records of variable size
Size	Variable

Certificate record contains a single LDS2-V Signer X.509 certificate data object. A Certificate Record MAY be referenced by one or more Visa Records.

The structure of the Certificate record in Visa Application is identical to the structure of the Certificate record in Travel Record Application specified in clause 3.2 Table 8.

Certificate records are written to EF.Certificates located under the Visa Records application DF using APPEND RECORD command. Certificate records can be read from EF.Certificates

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using READ RECORD command. Certificate records MUST NOT be updated or erased. The maximum number of records in EF.Certificates under the Visa Records application DF MUST be 254.

4.3 Application Selection

The Visa Records application MUST be selected by use of the Application Identifier (AID) as a reserved DF name. The AID MUST consist of the Registered Application Identifier assigned by ISO according to [ISO/IEC 7816-5] followed by the Proprietary Application Identifier Extension (PIX) of the Visa Records application:

- The Registered Application Identifier is 'A0 00 00 02 47'
- The Visa Records application MUST use PIX = '20 02';
- The full AID of the Visa Records application is 'A0 00 00 02 47 20 02'.

The IC MUST reject the selection of an application if the extension for this application is absent.

4.4 EF.VisaRecords (REQUIRED)

Visa Records MUST be stored in a single EF.VisaRecords having Linear Structure with Records of Variable Size.

Table 14: EF.VisaRecords

File Name	EF.VisaRecords
File ID	'0103'
Short EF Identifier	'03'
Select / FMM Access	PACE+TA (Visa record authorization bit b1 according to table 22)
Read Record / Search Record Access	PACE+TA (Visa record authorization bit b1 according to table 22)
Append Record Access	PACE+TA (Visa record authorization bit b2 according to table 22)
Write / Update Record Access	NEVER
Erase Record Access	NEVER
File structure	Linear structure with records of variable size
Size	Variable

Each Visa Record MUST contain a sequence of BER-TLV data objects (DO's 5F28 and 71), followed by the Authenticity Token (Signature) DO and DO containing reference to LDS2-V Signer certificate in EF.Certificates. Tag 71 contains a set of DO's (fields) listed in the table below.

Note: Interindustry tags specified in the table below are used in LDS context, so coexistent tag allocation scheme is not required.

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Table 15: EF.VisaRecords Format

Tag	Tag	Content	MANDATORY/ OPTIONAL/ CONDITIONAL	Format	Example
5F28		Issuing State or organization (Copy for SEARCH RECORD)	M	F (3) A	NLD
71	Visa Record (signed info)				
	5F28	Issuing State or organization	M	F (3) A	NLD
	43	Document Type	M	F (2) A,N,S	VS
	5F71	Machine-Readable Visa of Type A	O	F (48) A,N,S	
	5F72	Machine-Readable Visa of Type B	O	F (44) A,N,S	VCD<<DENT<<ARTHUR< PHILIP<<<<<<<<<<<
	5F73	Number of entries	O	V (1) B	'01' – 'FF'
	5F74	Duration of stay (days, months, years)	O	F (3) B	'010000' – 'FFFFFF'
	5F75	Passport number	O	F (9) A,N,S	XI85935F8
	5F76	Visa Type/class/category	O	V (4) B	
	5F77	Territory Information	O	V (8) B	
	49	Place of issuance (Issuing authority)	M	V (50) A,N,S	NEW YORK
	5F25	Effective Date (Date of issuance)	M	F (8) N	20120826 (yyyymmdd)
	5F24	Expiration Date	M	F (8) N	20130826 (yyyymmdd)
	5A	Document number	M	F (9) A,N,S	XI85935F8
	5F32	Additional information (endorsements: duration, limitations, and fees paid)	O	V (50) A,N,S	Free-form text
	5B	Name of holder (full name)	M	V (50) A,N,S	VAN DER STEEN MARIANNE LOUISE
	5F33	Primary Identifier (surname)	M	V (50) A,N,S	VAN DER STEEN
	5F34	Secondary Identifier (given name)	M	V (50) A,N,S	MARIANNE LOUISE
	5F35	Sex	M	F (1) A,S	F, M, or <
	5F2B	Date of birth	M	F (8) N,S	19870814 (yyyymmdd)
	5F2C	Nationality	M	F (3) A,S	NLD
	5F1F	MRZ	M	V (50) A,N,S	VAN<DER<STEEN<< MARIANNE<LOUISE
	5F40	Reference to Additional Biometrics EF	O	F (2) B	'0201'
5F37	Authenticity token (Signature)		M	V (140), B	'5F' '37' Len {Signature}
5F38	Reference (record number) to LDS2-V Signer certificate in Certificates Store		M	F (1) B	'01' ... 'FE'

Note 1: A = Alpha character [a..z, A..Z], N = Numeric character [0..9], S = Special character ['<' ' '], B = 8-bit Binary data (any other than A, N or S), F = fixed-length field, V = variable-length field.

The order of the data objects in a record is fixed. The IS **MUST** build up the record content using the data objects in the order specified in the table.

Each Visa Record MUST contain a digital signature (Authenticity Token) calculated over the DO'71, including Tag 71 and Length. Signature is generated by the LDS2-V Signer.

LDS2-V Signer certificates required to verify Visa Record's signature are stored in a separate EF.Certificates store located under the Visa Records application DF.

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Each Visa Record MUST be appended to EF.VisaRecords using APPEND RECORD. Visa Records and MUST NOT be altered (updated) or erased. The maximum number of records allowed in EF.VisaRecords MUST be 254.

Note 2: Issuing state 3-letter code according to ISO3166-1

Note 3: Optional DO'5F40, if present, MUST contain the 2 bytes identifier of the EF within the Additional Biometrics application containing biometric data. This DO may only be used provided the Additional Biometrics application is present on the eMRTD.

5. ADDITIONAL BIOMETRICS APPLICATION (CONDITIONAL)

The Additional Biometrics application MAY be implemented by an issuing State or organization. The following is conditionally REQUIRED if the optional Additional Biometrics application has been invoked or any visa record has referenced it.

5.1 File Structure Summary

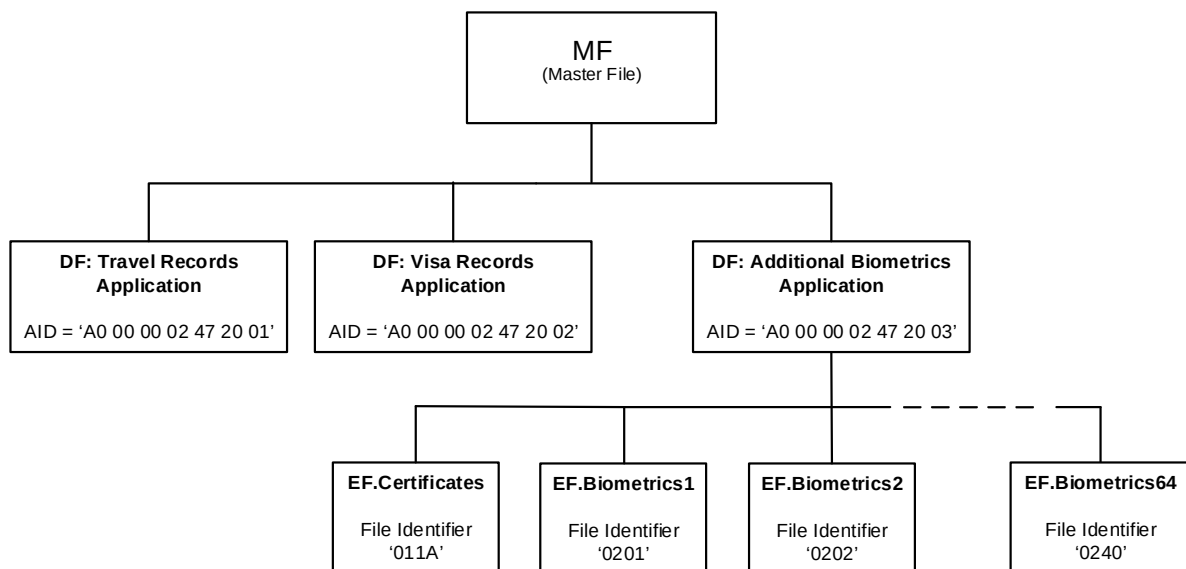


Figure 3: Additional Biometrics Application Structure

5.2 EF.Certificates (REQUIRED)

The Additional Biometrics Signer certificates are stored in EF.Certificates inside the application DF and having linear structure with records of variable size. These certificates are intended to be used by the IS to further offline validation of the digital signature in the EF.Biometrics.

Table 16: EF.Certificates

File Name	EF.Certificates
File ID	'011A'
Short EF Identifier	'1A'
Select / FMM Access	PACE+TA (Additional Biometrics authorization byte 1 bit b1 (see table 23))
Read Record/Search Record access	PACE+TA (Additional Biometrics authorization byte 1 bit b1 (see table 23))
Append Record Access	PACE+TA (Additional Biometrics authorization byte 1 bit b2 (see table 23))
Write / Update Record Access	NEVER
Erase Record Access	NEVER
File structure	Linear structure with records of variable size
Size	Variable

Certificate record contains a single Additional Biometrics Signer X.509 certificate data object. A Certificate Record MAY be referenced by one or more Additional Biometrics EF.

The structure of the Certificate record in Additional Biometrics Application is identical to the structure of the Certificate record in Travel Record Application specified in clause 3.2 Table 8.

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Certificate records are written to EF.Certificates located under the Additional Biometrics application DF using APPEND RECORD command. Certificate records can be read from EF.Certificates using READ RECORD command. Certificate records MUST NOT be updated or erased. The maximum number of records in EF.Certificates under the Additional Biometrics application DF MUST be 64.

5.3 Application Selection

The Additional Biometrics application MUST be selected by use of the Application Identifier (AID) as a reserved DF name. The AID MUST consist of the Registered Application Identifier assigned by ISO according to [ISO/IEC 7816-5] followed by the Proprietary Application Identifier Extension (PIX) of the Additional Biometrics application:

- The Registered Application Identifier is 'A0 00 00 02 47'
- The Additional Biometrics application MUST use PIX = '20 03';
- The full AID of the Additional Biometrics application is 'A0 00 00 02 47 20 03'.

The IC MUST reject the selection of an application if the extension for this application is absent.

5.4 EF.Biometrics

Additional Biometric MUST be stored under Additional Biometrics Application in Elementary Files having Transparent Structure as per [ISO/IEC 7816-4].

Each Additional Biometrics EF MAY be linked to one or more records in EF.VisaRecords in Visa Records Application (or other EFs and applications) using Additional Biometrics Elementary File Identifier.

Table 17: EF.Biometrics

File Name	EF.Biometrics
File ID	'0201' ... '0240'
Short EF Identifier	N/A
Select / FMM / Read Access in LCS Deactivated	PACE+TA (AdditionalBiometrics authorization according to table 23, bits b2, b4, b6, b8 of byte 2 - 17)
Write Access in LCS Deactivated	PACE+TA (AdditionalBiometrics authorization according to table 23, bits b2, b4, b6, b8 of byte 2 - 17)
Activate Access in LCS Deactivated	PACE+TA (AdditionalBiometrics authorization according to table 23, bits b2, b4, b6, b8 of byte 2 - 17)
Select / FMM / Read Access in LCS Activated	PACE+TA (AdditionalBiometrics authorization according to table 23, bits b1, b3, b5, b7 of byte 2 - 17)
Write Access in LCS Activated	NEVER
Activate Access in LCS Activated	NEVER
Erase Access	NEVER
File structure	Transparent structure
Size	Variable

Each Additional Biometric EF MUST contain a BER-TLV data object DO'7F2E encapsulating 3 data objects - the Biometric data DO'5F2E followed by the Authenticity Token (Signature) DO'5F37' and DO'5F38' containing the reference to an Additional Biometrics Signer certificate in EF.Certificates as shown in the table below.

The content of DO'5F2E is up to the Additional Biometrics issuer and out of scope of this specification.

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The Additional Biometrics EF creation mechanism is out of scope of this specification. Issuer SHOULD pre-create a number of Additional Biometrics EFs.

Note: Interindustry tags specified in the table below are used in LDS context, so coexistent tag allocation scheme is not required.

Table 18: EF.Biometrics Format

Tag	Tag	Content	MANDATORY/ OPTIONAL/ CONDITIONAL	Format	Example
7F2E		Biometric Data Template	M		'7F' '2E' Len {DO'5F2E' DO'5F37' DO'5F38'}
	5F2E	Additional Biometric data	M	V, B	'5F' '2E' Len {Biometric data}
	5F37	Authenticity token (Signature)	M	V (140), B	'5F' '37' Len {Signature}
	5F38	Reference (record number) to Additional Biometrics Signer certificate in Certificates Store	M	F (1) B	'01' ... '40'

Note 1: B = 8-bit Binary data (any other than A, N or S), F = fixed-length field, V = variable-length field.

The order of the data objects in EF is fixed.

Each Additional Biometrics EF MUST contain a digital signature (Authenticity Token) calculated over the DO'5F2E, including Tag and Length. Signature is generated by the Additional Biometrics Signer.

Additional Biometrics Signer certificate required to verify Additional Biometric's signature is stored in a separate EF.Certificates store located under the Additional Biometrics application DF.

Each Additional Biometrics EF MUST be written using WRITE BINARY command with option write-once (see 6.4.1).

Additional Biometrics EF MUST NOT be altered (updated) or erased. The maximum number of Additional Biometrics EFs is 64.

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All possible Additional Biometrics EF names, identifiers and short identifiers are listed in Table 19.

Table 19: EF.Biometrics identifiers

EF name	EF identifier	Short EF identifier	EF name	EF identifier	Short EF identifier
EF.Biometrics1	'0201'	N/A	EF.Biometrics33	'0221'	N/A
EF.Biometrics2	'0202'	N/A	EF.Biometrics34	'0222'	N/A
EF.Biometrics3	'0203'	N/A	EF.Biometrics35	'0223'	N/A
EF.Biometrics4	'0204'	N/A	EF.Biometrics36	'0224'	N/A
EF.Biometrics5	'0205'	N/A	EF.Biometrics37	'0225'	N/A
EF.Biometrics6	'0206'	N/A	EF.Biometrics38	'0226'	N/A
EF.Biometrics7	'0207'	N/A	EF.Biometrics39	'0227'	N/A
EF.Biometrics8	'0208'	N/A	EF.Biometrics40	'0228'	N/A
EF.Biometrics9	'0209'	N/A	EF.Biometrics41	'0229'	N/A
EF.Biometrics10	'020A'	N/A	EF.Biometrics42	'022A'	N/A
EF.Biometrics11	'020B'	N/A	EF.Biometrics43	'022B'	N/A
EF.Biometrics12	'020C'	N/A	EF.Biometrics44	'022C'	N/A
EF.Biometrics13	'020D'	N/A	EF.Biometrics45	'022D'	N/A
EF.Biometrics14	'020E'	N/A	EF.Biometrics46	'022E'	N/A
EF.Biometrics15	'020F'	N/A	EF.Biometrics47	'022F'	N/A
EF.Biometrics16	'0210'	N/A	EF.Biometrics48	'0230'	N/A
EF.Biometrics17	'0211'	N/A	EF.Biometrics49	'0231'	N/A
EF.Biometrics18	'0212'	N/A	EF.Biometrics50	'0232'	N/A
EF.Biometrics19	'0213'	N/A	EF.Biometrics51	'0233'	N/A
EF.Biometrics20	'0214'	N/A	EF.Biometrics52	'0234'	N/A
EF.Biometrics21	'0215'	N/A	EF.Biometrics53	'0235'	N/A
EF.Biometrics22	'0216'	N/A	EF.Biometrics54	'0236'	N/A
EF.Biometrics23	'0217'	N/A	EF.Biometrics55	'0237'	N/A
EF.Biometrics24	'0218'	N/A	EF.Biometrics56	'0238'	N/A
EF.Biometrics25	'0219'	N/A	EF.Biometrics57	'0239'	N/A
EF.Biometrics26	'021A'	N/A	EF.Biometrics58	'023A'	N/A
EF.Biometrics27	'021B'	N/A	EF.Biometrics59	'023B'	N/A
EF.Biometrics28	'021C'	N/A	EF.Biometrics60	'023C'	N/A
EF.Biometrics29	'021D'	N/A	EF.Biometrics61	'023D'	N/A
EF.Biometrics30	'021E'	N/A	EF.Biometrics62	'023E'	N/A
EF.Biometrics31	'021F'	N/A	EF.Biometrics63	'023F'	N/A
EF.Biometrics32	'0220'	N/A	EF.Biometrics64	'0240'	N/A

6. FILE ACCESS CONDITIONS

6.1 Roles and Default Authorization Levels (REQUIRED)

Each CV certificate contains a Certificate Holder Authorization Template (CHAT) that identifies the certificate holder role (IS, DV, CVCA) and contains access rights to DG3/DG4 of the REQUIRED eMRTD Application (for legacy reasons or other national uses).

CHAT comprises a sequence of 2 objects:

1. An object identifier specifying the terminal type and the format of the template [TR-03110]:
 id-roles OBJECT IDENTIFIER ::= {bsi-de applications(3) mrttd(1) 2}
 id-IS OBJECT IDENTIFIER ::= {id-roles 1}
2. A discretionary data object (tag '53') containing bit-encoded role and read-only access rights of the certificate holder according to the following table:

Table 20: Default CHAT Authorization

	Description	Byte 1							
		b8	b7	b6	b5	b4	b3	b2	b1
Role	CVCA	1	1						
	DV (domestic)	1	0						
	DV (foreign)	0	1						
	IS	0	0						
Read Access	RFU								
	RFU								
	RFU								
	RFU								
	DG4 (Iris)							1	
	DG3 (Finger)								1

Note: The eMRTD MUST ignore the value of RFU bits in the Certificate Holder Authorization.

6.2 Application Authorization Levels (REQUIRED)

Certificate holder authorizations for each LDS2 application are encoded in CV-certificate-extensions (one extension per application). Certificate extension is a discretionary template ('73') comprising 2 data objects - an Authorization Object Identifier (tag '06') for a specific application and a discretionary data object (tag '53') containing bit-encoded access rights of the certificate holder to specified application.

To determine the effective authorization of a certificate holder, the eMRTD chip calculates a bitwise Boolean 'and' of the access rights contained in the certificate extensions of the IS Certificate and referenced DV and CVCA Certificates.

For Travel Records application the Authorization Object Identifiers and access right encoding are:

id-icao-lds2-travelRecords OBJECT IDENTIFIER ::= {id-icao-lds2 1}
 id-icao-lds2-travelRecords-access OBJECT IDENTIFIER ::= {id-icao-lds2-travelRecords 3}

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Table 21: Authorizations for Travel Records Application

	Description	Byte 1							
		b8	b7	b6	b5	b4	b3	b2	b1
Access rights	RFU								
	RFU								
	RFU								
	RFU								
	Append EF.Certificates					1			
	Read/Search/Select/FMM EF.Certificates						1		
	Append EF.EntryRecords/ExitRecords							1	
	Read/Search/Select/FMM EF.EntryRecords/ExitRecords								1

For Visa Records application the Authorization Object Identifiers and access right encoding are:

id-icao-lds2-visaRecords OBJECT IDENTIFIER ::= {id-icao-lds2 2}
id-icao-lds2-visaRecords-access OBJECT IDENTIFIER ::= {id-icao-lds2-visaRecords 3}

Table 22: Authorizations for Visa Records Application

	Description	Byte 1							
		b8	b7	b6	b5	b4	b3	b2	b1
Access rights	RFU								
	RFU								
	RFU								
	RFU								
	Append EF.Certificates				1				
	Read/Search/Select/FMM EF.Certificates					1			
	Append EF.VisaRecords						1		
	Read/Search/Select/FMM EF.VisaRecords								1

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For Additional Biometrics application the Authorization Object Identifiers and access right encoding are:

id-icao-lds2-additionalBiometrics OBJECT IDENTIFIER ::= {id-icao-lds2 3}

id-icao-lds2-additionalBiometrics-access OBJECT IDENTIFIER ::= {id-icao-lds2-additionalBiometrics 3}

Table 23: Authorizations for Additional Biometrics Application

	Description	EF identifier	Authorizations							
			b8	b7	b6	b5	b4	b3	b2	b1
Byte 1	RFU									
	RFU									
	RFU									
	RFU									
	RFU									
	RFU									
Byte 2	Append EF.Certificates	011A							1	
	Select/FMM/Read/Search EF.Certificates	011A								1
	Select/FMM/Write/Activate/Read EF.Biometrics1 in Deactivated LCS	0201	1							
	Select/FMM/Read EF.Biometrics1 in Activated LCS	0201		1						
	Select/FMM/Write/Activate/Read EF.Biometrics2 in Deactivated LCS	0202			1					
	Select/FMM/Read EF.Biometrics2 in Activated LCS	0202				1				
	Select/FMM/Write/Activate/Read EF.Biometrics3 in Deactivated LCS	0203					1			
	Select/FMM/Read EF.Biometrics3 in Activated LCS	0203						1		
	Select/FMM/Write/Activate/Read EF.Biometrics4 in Deactivated LCS	0204							1	
	Select/FMM/Read EF.Biometrics4 in Activated LCS	0204								1
...										
Byte 17	Select/FMM/Write/Activate/Read EF.Biometrics61 in Deactivated LCS	023D	1							
	Select/FMM/Read EF.Biometrics61 in Activated LCS	023D		1						
	Select/FMM/Write/Activate/Read EF.Biometrics62 in Deactivated LCS	023E			1					
	Select/FMM/Read EF.Biometrics62 in Activated LCS	023E				1				
	Select/FMM/Write/Activate/Read EF.Biometrics63 in Deactivated LCS	023F					1			
	Select/FMM/Read EF.Biometrics63 in Activated LCS	023F						1		
	Select/FMM/Write/Activate/Read EF.Biometrics64 in Deactivated LCS	0240							1	
	Select/FMM/Read EF.Biometrics64 in Activated LCS	0240								1

Note 1 - See table 17 for a mapping of the authorizations to the EF.Biometrics life cycle state.

Note 2 - The eMRTD MUST ignore the value of RFU bits in the Certificate Holder Authorization.

Note 3 - Issuing State or organization MUST NOT issue terminal certificates with Write/Activate authorizations to the IS that are only supposed to read Additional Biometrics and not supposed to write them.

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6.3 Records Handling

Travel Records, Visa Records and Certificates MUST be stored in EF under the respective applications and having Linear Structure with Records of Variable Size.

Records within each EF MUST be referenced by a record number. Each record number MUST be unique and sequential (zero is out of the scope of this specification).

Within each EF supporting a linear structure, the record numbers MUST be sequentially assigned when appending, such as in the order of creation; the first record (number one) is the first created record.

The following [ISO/IEC 7816-4] commands MUST be used for records access:

- APPEND RECORD Writing/appending of Travel Records, Visas, Certificates
- READ RECORD Reading of one or more Travel Records, Visas, Certificates
- SEARCH RECORD Search of one or more Travel Records, Visas, Certificates

Note: Acronyms used in this sub-clause are defined in ISO/IEC 7816-4.

6.3.1 APPEND RECORD command

The command initiates the writing of a new record at the end of a linear structure.

Table 24: APPEND RECORD Command

CLA	'00' / '0C'
INS	'E2'
P1	'00' (any other value is invalid)
P2	See Table 26
L _c field	Length of the command data field
Data field	Record to be appended
L _e field	Absent

Table 25: APPEND RECORD Response

Data field	Absent
SW1-SW2	'9000', Checking or Execution error '6A84' Not enough memory space in the file '67 00' Wrong length (the record to be appended is longer than the specified maximum length)

Table 26: Coding of P2 in the APPEND RECORD Command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
X	X	X	X	X	-	-	-	Short EF identifier
-	-	-	-	-	0	0	0	Any other value is RFU

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6.3.2 READ RECORD Command

The response data field returns the contents of the addressed specified record(s) [or the beginning part of one record] within the respective EF.

Figure 4 illustrates the response data field. The comparison of N_r with the TLV structure indicates whether the unique record (read one record) or the last record (read all records) is incomplete, complete or padded.

Table 27: READ RECORD Command

CLA	'00' / '0C'	
INS	'B2'	
P1	Record number ('00' references the current record)	
P2	See Table 29	
L _c field	Absent	
Data field	INS = 'B2'	Absent
L _e field	Maximum number of bytes to be read encoded as extended length field; Le = '00 00 00' (any other value is out of the scope of the specification)	

Table 28: READ RECORD Response

Data field	Data read
SW1-SW2	'9000', Checking or Execution error, '6A83' (Record not found)

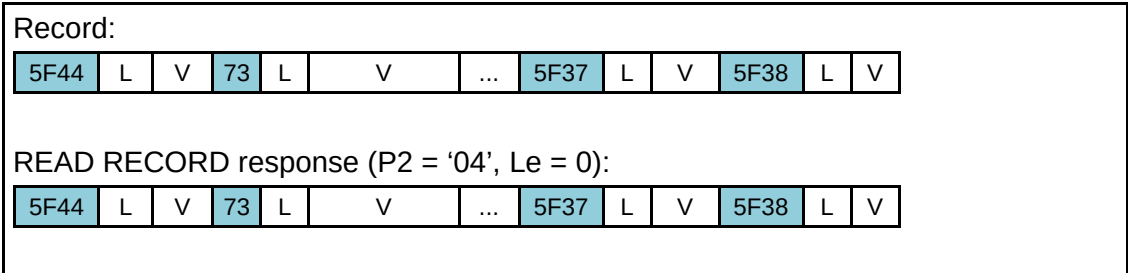
Table 29 - Coding of P2 with the READ RECORD Command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	x	x	x	-	-	-	Short EF identifier
-	-	-	-	-	1	x	x	Record number in P1
-	-	-	-	-	1	0	0	— Read record P1
-	-	-	-	-	1	0	1	— Read all records from P1 up to the last
					1	1	1	RFU

If the L_e field contains only bytes set to '00', then the command should read completely either the single requested record, or the requested sequence of records, depending on bits 3, 2 and 1 of P2 and within the limit of maximum supported length for extended L_e field.

Note: The READ RECORD command with short length fields is out of the scope of this specification

Case a — Complete read of one record (the Le field contains only bytes set to '00')



Case b — Read several records up to the file end (the Le field contains only bytes set to '00')

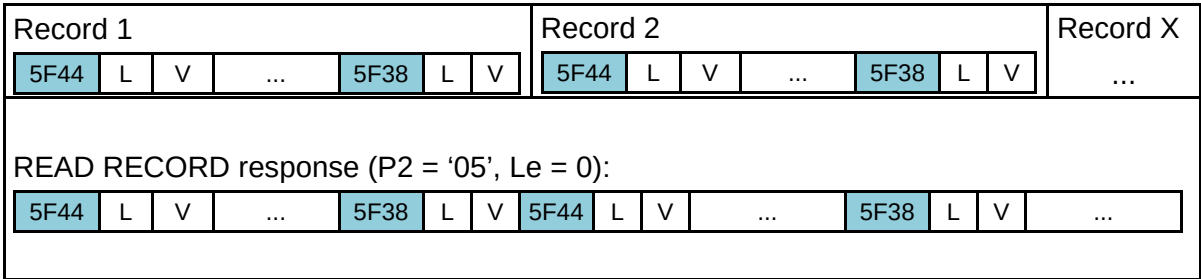


Figure 4: Response data fields

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6.3.3 SEARCH RECORD Command

The command initiates an enhanced multiple records search on records stored within the respective EF. In case of the enhanced multiple records search option the data field contains Record handling DO'7F76' defining the search window - offset within the record and the number of bytes to compare. The response data field returns the Record handling DO'7F76' containing one or more DO'02' containing number of record matching the search criteria within the designated EF. In case of the enhanced multiple records search option the command MUST set the first record matching the search criteria as current record.

In an EF supporting records of variable size with linear structure, the search MAY NOT take into account the records with a search window shorter than the search string.

Table 30: SEARCH RECORD Command

CLA	'00' / '0C'
INS	'A2'
P1	'00'
P2	See Table 32
L _c field	Length of command data field
Data field	Record handling DO'7F76' (See Table 33)
L _e field	'00' (short length) or '00 00' (extended length)

Table 31: SEARCH RECORD Response

Data field	Record handling template DO'7F76' containing one file reference DO'51' with one or more integer DO'02' containing record number matching the search criteria
SW1-SW2	'9000', Checking or Execution error or Warning '6282' (Unsuccessful search)

Note 1 - The response data field may be absent if no match is found.

Table 32: Coding of P2 for the SEARCH RECORD Command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
1	1	1	1	1	0	0	0	Multiple record handling
- Any other value is RFU.								

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Table 33: Record handling template for enhanced multiple record search

Tag	Value		Notes
'7F76'			Record handling DO
	Tag	Value	
	'51'	File identifier or short EF identifier	File reference DO
	'A1'	Search configuration template	
	Tag	Value	
	'80'	'00' / '30'	Search configuration parameter: - search in record number ascending order - step-width for the search: byte-wise - search termination: '00' – Search all addressed records '30' - Terminate search after first matching
	'B0'	Search window template	
	Tag	Value	
	'02'	Offset	
	'02'	Number of bytes	
	Tag	Value	
	'A3'	Search string template	
	Tag	Value	
	'B1'		
	Tag	Value	
	'81'	Search string	

Note 1: An empty offset DO in the search window template is not supported.

Note 2: If the search window template makes use of the value '00' for the number of bytes, the eMRTD chip MUST search all bytes from the offset in the records.

Note 3: The SEARCH RECORD command supports only the DOs specified in table 33. This implies that the SEARCH RECORD command supports exactly 1 file reference DO in the record handling DO and exactly 1 search string in the search string template. The command MAY ignore additional DOs or answer with an error code if additional DOs are used.

6.4 Transparent Files Handling

The Additional Biometrics transparent EFs are created by the eMRTD issuer in Operational Deactivated state (creation mechanism is out of scope of this specification). In Deactivated state the EF can be selected, written, updated and read with appropriate authorizations. See 6.2, Tables 17 and 23 for details on authorizations.

The following [ISO/IEC 7816-4] commands MUST be used for writing and reading Additional Biometrics transparent EF:

- UPDATE BINARY Writing of Additional Biometrics (Table 34)
- READ BINARY Reading of Additional Biometrics

The following [ISO/IEC 7816-9] command MUST be used for activating the transparent EF after writing and optional reading and verification are successfully finished:

- ACTIVATE Activating of Additional Biometrics EF

In Activated state the EF can be selected and read with appropriate authorizations (related to the Activated state), but can't be written (appended or updated) with any authorization.

The File and Memory Management (FMM) command MUST be used before writing to determine if there is enough available memory space in the EF.

The IS MUST use the following writing sequence for the EF.Biometrics:

- 1) The first UPDATE BINARY (odd INS) command MUST contain the following DO's in the data field:
 - DO'54 containing the offset '00';
 - DO'53 which MAY contain the first block of the EF. This DO MAY be empty ('53 00');
 - Proprietary DO'C0 indicating the total EF size (memory size to allocate);

The eMRTD MAY use the EF size information in DO'C0 for the internal memory allocation (e.g. for explicit dynamic memory allocation). If the eMRTD doesn't support the EF size information DO (ex., memory has been allocated statically by the issuer, or eMRTD supports implicit dynamic EF memory reallocation), then the eMRTD MAY ignore the DO'C0, proceed with writing of the first block of the EF and return '9000', or it MAY return the '6A80' (incorrect parameter in the command data field) error.

If the eMRTD returns any error in response to UPDATE BINARY with the proprietary DO'C0, then the IS MUST send the standard [ISO/IEC 7816-4] UPDATE BINARY (odd INS) command with zero offset DO'54 and DO'53, without the DO'C0.

- 2) Subsequent UPDATE BINARY (odd INS, without DO'C0) commands SHOULD use the offset n+1 where n denotes the number of bytes written so far to the EF.Biometrics. I.e. the terminal SHOULD sequentially write the EF data without a gap or overlap between the two consecutive UPDATE BINARY commands.
- 3) READ BINARY command MAY be used after any UPDATE BINARY command to verify the data written to the EF.
- 4) The ACTIVATE command MUST finalize EF.Biometrics personalization by permanently disabling writing into the EF.

6.4.1 UPDATE BINARY Command

A contactless IC which supports the Additional Biometrics Application MUST support the UPDATE BINARY command with odd INS byte 'D7' according to the Table 34.

The value of the BER-TLV Offset Data Object in the command data field specifies the offset; the value of the BER-TLV Discretionary Data Object in the command data field specifies the data to be written; the value of the optional BER-TLV File Size Data Object in the command data field specifies the total EF size. The length fields of these BER-TLV data objects should be encoded as short as possible.

Table 34: UPDATE BINARY Command with odd INS

CLA	'0C' - '8C'
INS	'D7'
P1	File identifier
P2	'00 00' identifies the current EF
Lc	Length of the command data field
Data	Offset Data Object (tag '54') Discretionary Data Object (tag '53') File Size Data Object (tag 'C0') (optional)
Le	absent

Table 35: UPDATE BINARY Response

Data field	Absent
SW1- SW2	'9000', Checking or Execution error '6A84' (Not enough memory space in the file) '6A80' Incorrect parameters in the command data field (ex., DO'C0 not supported) '6982' Security status not satisfied: The EF.Biometrics is in EF LCS Activated

If the Inspection System does not follow the UPDATE BINARY sequence as specified in clause 6.4 (i.e. the first UPDATE BINARY does not start at offset 0), the eMRTD chip MAY terminate the UPDATE BINARY command with an error.

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6.4.2 ACTIVATE Command

The ACTIVATE command initiates the transition of the currently selected Additional Biometrics EF from the Deactivated life cycle state (LCS) to the Activated LCS.

Table 36: ACTIVATE Command

CLA	'0C'
INS	'44'
P1	'00'
P2	'00'
Lc	Absent
Data field	Absent
Le	Absent

Table 37: ACTIVATE Response

Data field	Absent
SW1-SW2	'9000', Checking or Execution error

After successful execution of this command, the currently selected EF.Biometrics MUST be switched to the Activated LCS. In case an error occurs (SW different from '9000'), the currently selected EF.Biometrics MUST remain in the Deactivated state.

Immediately after successful execution of this command (Status Word = '9000'), the effective authorization required to perform an action on the EF.Biometrics MUST be the one corresponding to the Activated state (according to the table 17). The effective authorization corresponding to the Deactivated state MUST NOT raise any access right on the EF.Biometrics anymore.

6.5 Memory Management

6.5.1 File and Memory Management Command

FILE AND MEMORY MANAGEMENT (FMM) command initiates a query of the used or free memory size for the addressed EF. This command is provided for eMRTD as proprietary. This command may be used for checking the available free space for the addressed EF before writing or appending. Also this command may be used for getting the last appended record number for reading. P1 indicates the EF addressing method - current EF or file reference DO'51' can be used. P2 indicates the content of the query. The total number of bytes in the addressed EF with transparent or record structure and the number of existing or remaining records for the addressed record EF are provided. The total number of bytes comprises bytes available in the EF without any structural information. This number excludes any structural information that may be required by the eMRTD chip. The assumption for the number of remaining records is that the size of all remaining records is maximum. After a successful FMM command, the referenced EF becomes the current EF.

Table 38: FILE AND MEMORY MANAGEMENT (FMM) command

CLA	'8C'	Proprietary command with secure messaging
INS	'5F'	FILE AND MEMORY MANAGEMENT
P1	See Table 39	
P2	See Table 40	
Lc	Absent for encoding Nc = 0, present for encoding Nc > 0	
Data field	P1 = '00'	Absent
	P1 = '01'	File reference DO'51' (See Table 7 in ISO/IEC 7816-4:2013)
Le	'00'	Expected size of response data field

Data field	Absent or file and memory management DO'7F78'
SW1-SW2	See Table 5 and Table 6 in ISO/IEC 7816-4:2013

P1 specifies the EF selection method. P2 contains a bit mask specifying which information MUST be included in the response.

Table 39: Coding of P1 in the FFM command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
0	0	0	0	0	0	0	0	Current EF
0	0	0	0	0	0	0	1	File reference DO'51 in the command data field
- Any other value is RFU.								

Table 40: Coding of P2 in the FFM command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
-	-	-	-	-	-	-	1	Total number of bytes in the addressed EF
-	-	-	-	-	-	1	-	Number of remaining records in the addressed record EF
-	-	-	-	-	1	-	-	Number of existing records in the addressed record EF
x	x	x	x	x	-	-	-	0000 (any other value is RFU)
- Any other value is RFU.								

Table 41: Coding of DO'51 in the FMM command data field

Tag	Length	Value
'51'		

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	1	Short EF identifier (bits b8 to b4 encode a number from one to thirty; bits b3 to b1 are set to 000)
	2	EF identifier

The FMM command response contains a set of DO's representing requested file and memory size information.

Table 42: FMM command response

Data field	Absent or control information according to P2. See Table 43.
SW1-SW2	'9000', Checking or Execution error as per Table 6 in [ISO/IEC 7816-4]

Table 43: File and Memory management

Tag	Length	Value		
'7F78'	Var.	File and memory management DOs		
		Tag	Len	Value
		81	Var	Total number of bytes in the addressed EF
		82	Var	Number of remaining records in the addressed record EF
		83	Var	Number of existing records in the addressed record EF

Note 1: The eMRTD chip MUST return only the Data objects in the file and memory management DO that are requested by means of P2.

Note 2: The FMM response data is valid only for the specified EF. FMM response data from different EFs may not be independent, e.g. if different EFs share the available memory. The IS should take this into account if combining FMM response data of different EFs.

Note 3: Secure Messaging tag '85' MUST be used for the FMM command data.

6.6 Object Identifiers

6.6.1 Legacy and New Application Object Identifiers Summary

Table 44: LDS1.7, LDS1.8 and LDS2 OIDs

Object Identifier	Value	Comments
id-icao	2.23.136	ICAO OID
id-icao-mrtd	id-icao 1	eMRTD OID
id-icao-mrtd-security	id-icao-mrtd 1	
id-icao-ldsSecurityObject	id-icao-mrtd-security 1	LDS security object
id-icao-mrtd-security-cscaMasterList	id-icao-mrtd-security 2	CSCA master list
id-icao-mrtd-security-cscaMasterListSigningKey	id-icao-mrtd-security 3	
id-icao-mrtd-security-documentTypeList	id-icao-mrtd-security 4	document type list
id-icao-mrtd-security-aaProtocolObject	id-icao-mrtd-security 5	Active Authentication protocol
id-icao-mrtd-security-extensions	id-icao-mrtd-security 6	CSCA name change
id-icao-mrtd-security-extensions-nameChange	id-icao-mrtd-security-extensions 1	
id-icao-mrtd-security-extensions-documentTypeList	id-icao-mrtd-security-extensions 2	DS document type
id-icao-DeviationList	id-icao-mrtd-security 7	Defect List Base OIDs
id-icao-DeviationListSigningKey	id-icao-mrtd-security 8	
id-icao-lds2	id-icao-mrtd-security 9	LDS2 Object Identifiers
id-icao-lds2-travelRecords	id-icao-lds2 1	Travel Records application base OID
id-icao-lds2-travelRecords-application	id-icao-lds2-travelRecords 1	Travel Records AID
id-icao-lds2-travelRecords-signing	id-icao-lds2-travelRecords 2	LDS2-TS signer certificate
id-icao-lds2-travelRecords-access	id-icao-lds2-travelRecords 3	Authorization certificate extension
id-icao-lds2-visaRecords	id-icao-lds2 2	Visa Records application base OID
id-icao-lds2-visaRecords-application	id-icao-lds2-visaRecords 1	Visa Records AID
id-icao-lds2-visaRecords-signing	id-icao-lds2-visaRecords 2	LDS2-V signer certificate
id-icao-lds2-visaRecords-access	id-icao-lds2-visaRecords 3	Authorization certificate extension
id-icao-lds2-additionalBiometrics	id-icao-lds2 3	Additional Biometrics base OID
id-icao-lds2-additionalBiometrics-application	id-icao-lds2-additionalBiometrics 1	Additional Biometrics AID
id-icao-lds2-additionalBiometrics-signing	id-icao-lds2-additionalBiometrics 2	LDS2-B signer certificate
id-icao-lds2-additionalBiometrics-access	id-icao-lds2-additionalBiometrics 3	Authorization certificate extension
id-icao-spoc	id-icao-mrtd-security 10	SPOC Object Identifiers
id-icao-spocClient	id-icao-spoc 1	Client
id-icao-spocServer	id-icao-spoc 2	Server

6.6.2 ASN.1 Specifications

-- LDS2 Travel Records application Object Identifiers

id-icao-lds2-travelRecords	OBJECT IDENTIFIER	::= {id-icao-lds2 1}
id-icao-lds2-travelRecords-application	OBJECT IDENTIFIER	::= {id-icao-lds2-travelRecords 1}
id-icao-lds2-travelRecords-signing	OBJECT IDENTIFIER	::= {id-icao-lds2-travelRecords 2}
id-icao-lds2-travelRecords-access	OBJECT IDENTIFIER	::= {id-icao-lds2-travelRecords 3}

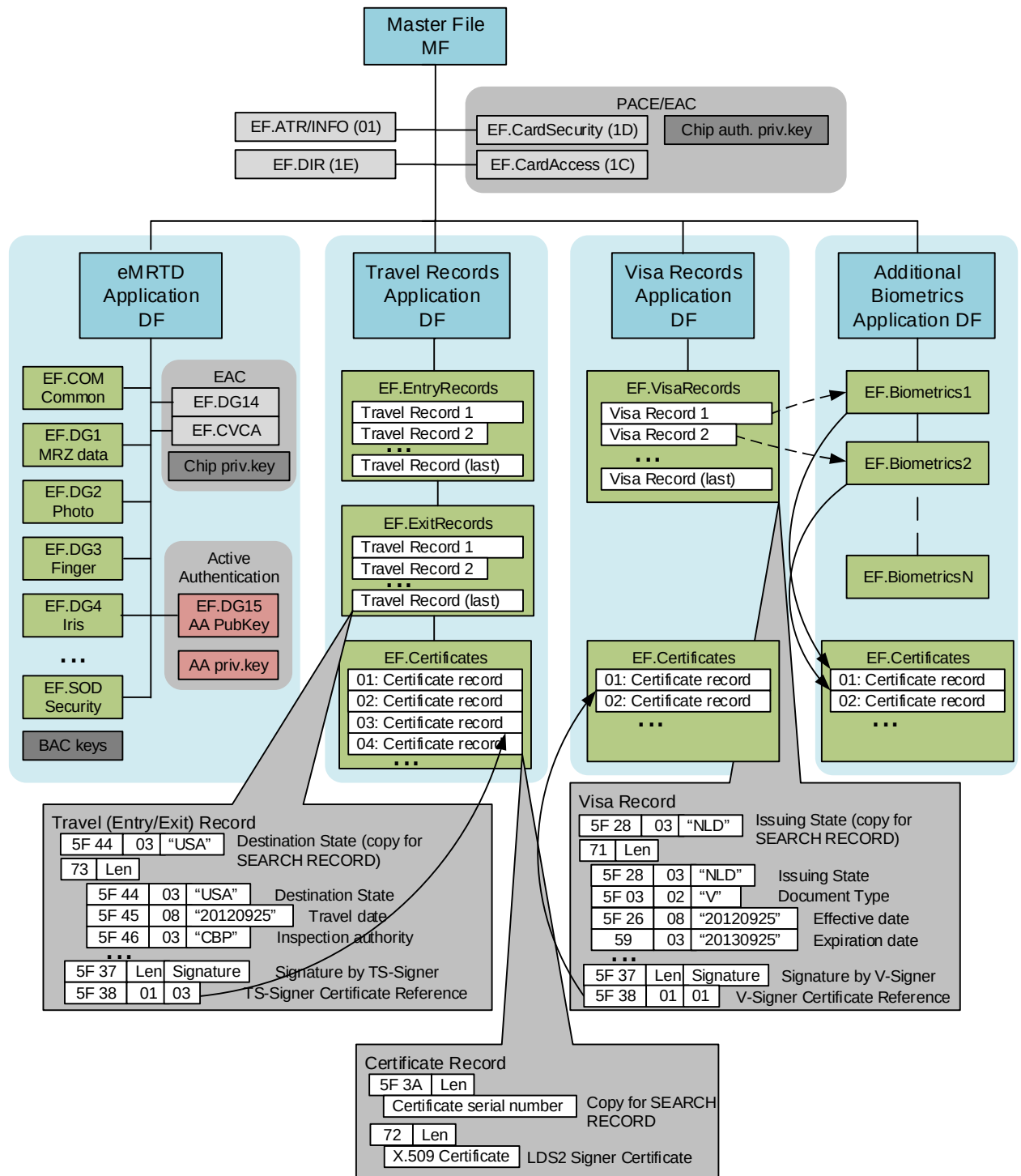
-- LDS2 Visa Records application Object Identifiers

id-icao-lds2-visaRecords	OBJECT IDENTIFIER	::= {id-icao-lds2 2}
id-icao-lds2-visaRecords-application	OBJECT IDENTIFIER	::= {id-icao-lds2-visaRecords 1}
id-icao-lds2-visaRecords-signing	OBJECT IDENTIFIER	::= {id-icao-lds2-visaRecords 2}
id-icao-lds2-visaRecords-access	OBJECT IDENTIFIER	::= {id-icao-lds2-visaRecords 3}

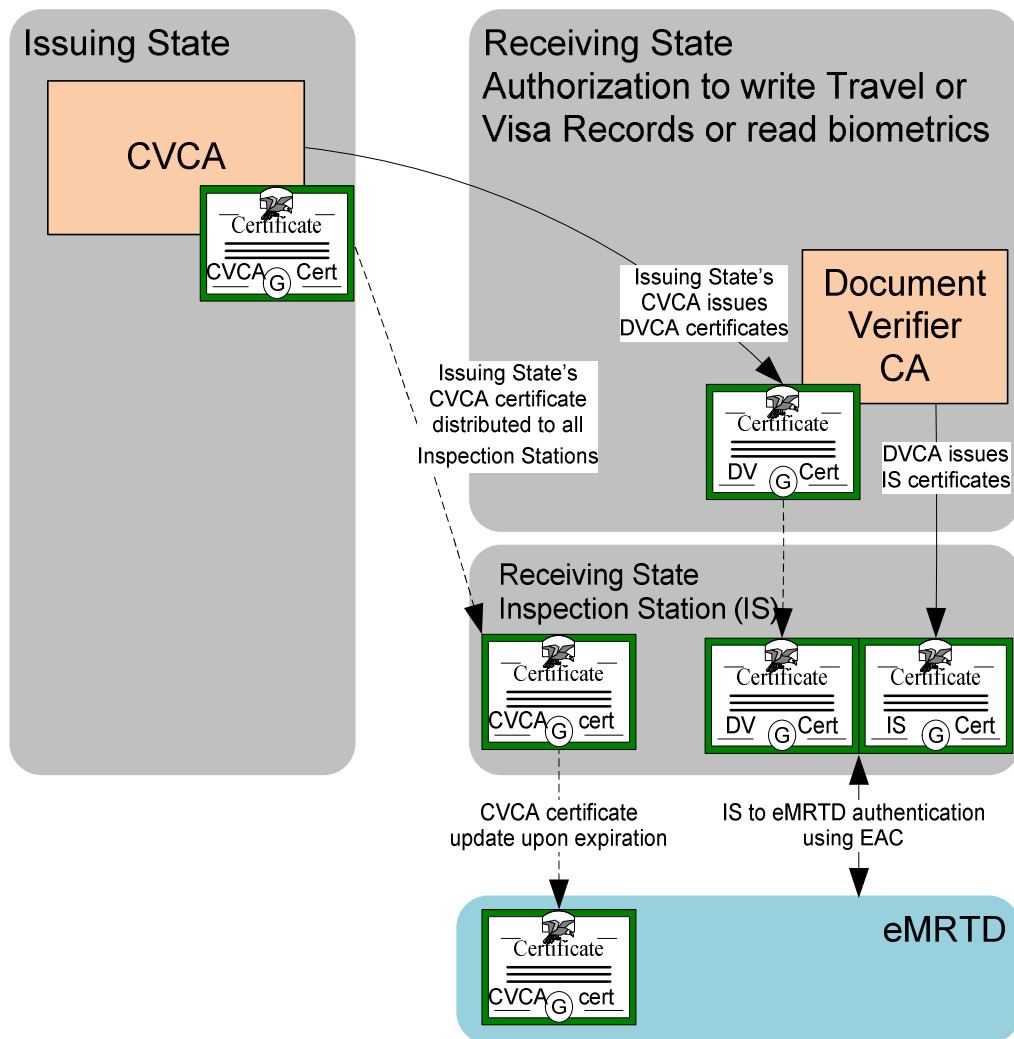
-- LDS2 Additional Biometrics application Object Identifiers

id-icao-lds2-additionalBiometrics	OBJECT IDENTIFIER	::= {id-icao-lds2 3}
id-icao-lds2-additionalBiometrics-application	OBJECT IDENTIFIER	::= {id-icao-lds2-additionalBiometrics 1}
id-icao-lds2-additionalBiometrics-signing	OBJECT IDENTIFIER	::= {id-icao-lds2-additionalBiometrics 2}
id-icao-lds2-additionalBiometrics-access	OBJECT IDENTIFIER	::= {id-icao-lds2-additionalBiometrics 3}

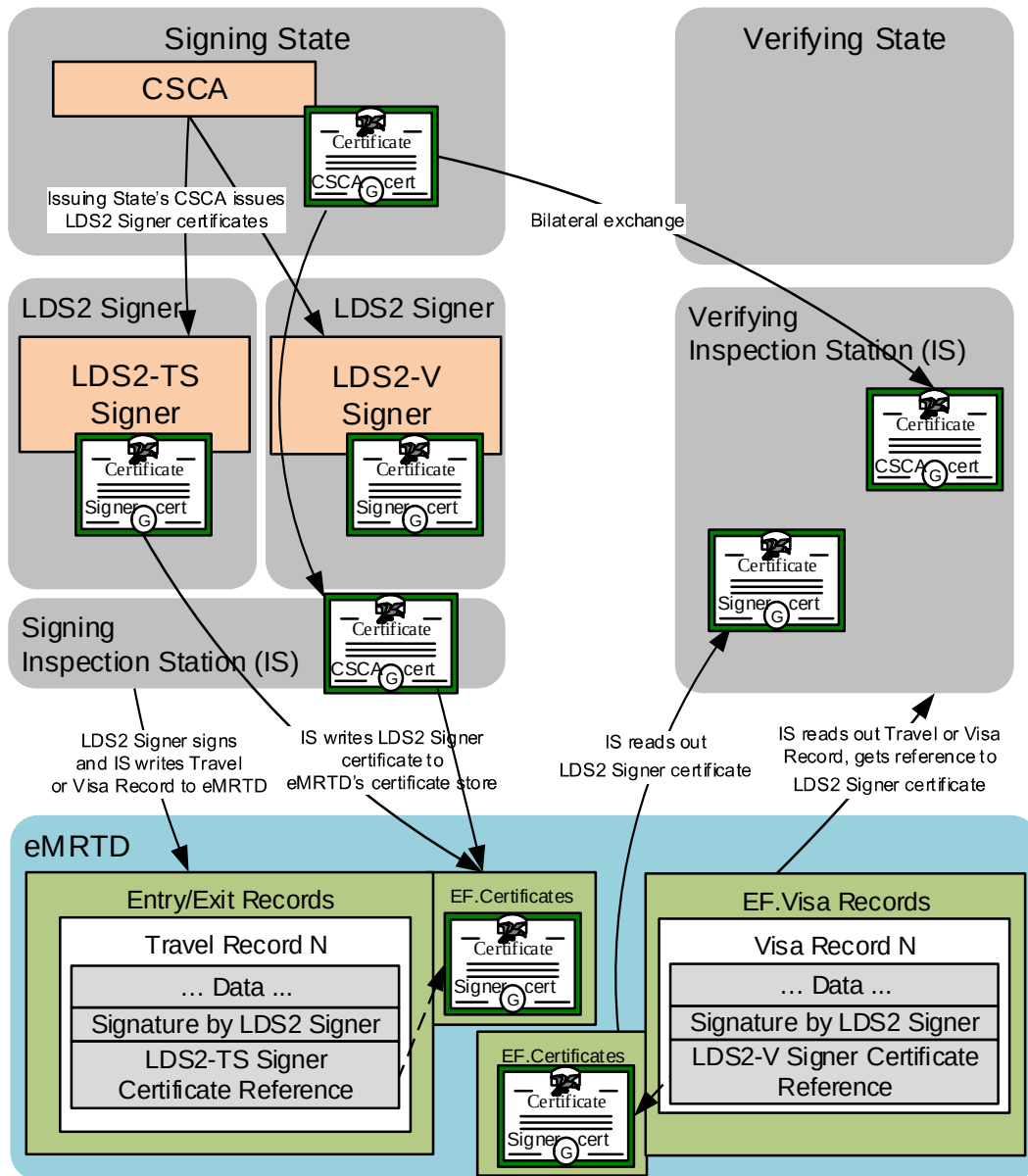
Annex A FILE STRUCTURES SUMMARY



Annex B LDS AUTHORIZATION SUMMARY



Annex C LDS DIGITAL SIGNATURE SUMMARY



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Annex D EXAMPLE READING TRAVEL RECORDS

1) FMM command retrieving the number of Entry Records

CLA	INS	P1	P2	Lc	Data	Le
8C	5E	01	08	04	51 02 01 01	00

CLA: Interindustry class / no secure messaging

INS: FMM

P1: 01 - EF identifier in command data field

P2: 10 - Return existing number of records in a record EF

Lc: 04

Data: DO'51 containing Entry Records EF identifier '0101'

Le : 00 (Short Le)

Response: File and Memory Management DO representing the number of records in the EF

Data	SW1-SW2
7F78 03 84 01 FD	90 00

The DO in the response data contains the last record number which can be used in the next READ RECORD command (P1).

Ex., last record number '00' means that there are no records in this file, response 'FD' means that number of records is 253 (maximum number of records is 254).

2a) READ RECORD command retrieving the last Travel Record from the retrieved list

The following command can be used to retrieve a single record using record number returned by the FMM command:

CLA	INS	P1	P2	Le
00	B2	FD	04	00 00 00

CLA : Interindustry class / no secure messaging

INS : READ RECORD(S)

P1 : Record number from the previous command's response

P2 : Record number in P1 / read record P1

Le : 00 00 00 (Extended Le) - read entire record

Response: Record - 253 (0xFD)

Data	SW1-SW2
5F44 Len <Data> 73 Len <Data> 5F37 Len <Data> 5F38 Len <Data>	90 00

2b) READ RECORD retrieving last 2 Travel Records from retrieved the list

The following command can be used to retrieve 2 (or more) records from the list returned by FMM command. Reading several records in one APDU exchange improves performance. The number of records that can be retrieved by a single command can be determined from extended length information in EF.ATR/INFO and maximum size of Travel Record.

CLA	INS	P1	P2	Le
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00	B2	FC	05	00 00 00
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CLA : Interindustry class / no secure messaging

INS : READ RECORD(S)

P1 : Decremental Record number from the FMM response (253 - 1 = 252 = 'FC')

P2 : Record number in P1 / read all records from P1 up to the last

Le : 00 00 00 (Extended Le) - read entire record

Response: Last 2 records - 252 (0xFC) and 253 (0xFD)

Data	SW1-SW2
5F44 Len <Data> 73 Len <Data> 5F37 Len <Data> 5F38 Len <Data> 5F44 Len <Data> 73 Len <Data> 5F37 Len <Data> 5F38 Len <Data>	90 00

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Annex E EXAMPLE SEARCHING RECORDS BY STATE

1) SEARCH RECORD command searching Travel Record(s) by Destination State

CLA	INS	P1	P2	Lc	Data	Le
00	A2	00	F8	Var	7F 76 Len 51 01 01 A1 0B 80 01 00 B0 06 02 01 03 02 01 03 A3 07 B1 05 81 03 xx xx xx	00

CLA: Interindustry class / no secure messaging

INS: SEARCH RECORD(S)

P1: record number = 00

P2: Search through multiple EFs

Lc: length of command data field

Data: DO'7F76' - Record handling DO

DO'51' - File reference DO (EF.EntryRecords short identifier '01')

DO'A1' - Search configuration template

DO'80' - Search configuration parameter: '00' (search all records)

DO'B0' - Search window template

DO'02' - Offset: '03'

DO'02' - Number of bytes: '03'

DO'A3' - Search string template

DO'B1' - Search string DO

DO'81' - Search string (country code): xx xx xx

Le: 00 (Short Le)

Response: DO'7F76' – Record handling DO

DO'51' - EF.EntryRecords short identifier '01'

One or more DO'02' containing matching record numbers

Data	SW1-SW2
7F 76 Len 51 01 01 02 01 03 02 01 04	90 00

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Annex F EXAMPLE WRITING TRAVEL RECORD AND CERTIFICATE

1) SEARCH RECORD command searching EF.Certificates by a Certificate Serial Number

IS checks if LDS2-TS Signer certificate with required serial numbers exists in EF.Certificates. The following command can be used for searching certificates:

CLA	INS	P1	P2	Lc	Data	Le
00	A2	00	F8	Var	7F 76 Len 51 01 1A A1 0B 80 01 00 B0 06 02 01 03 02 01 {Search string size} A3 Len B1 Len 81 Len xx xx .. xx xx	00

CLA: Interindustry class / no secure messaging

INS: SEARCH RECORD(S)

P1: record number = 00

P2: Search through multiple EFs

Lc: length of command data field

Data: DO'7F76' - Record handling DO

DO'51' - File reference DO (EF.Certificates short identifier '1A')

DO'A1' - Search configuration template

DO'80' - Search configuration parameter: '30' (stop if record found)

DO'B0' - Search window template

DO'02' - Offset: '03'

DO'02' - Number of bytes: Search string size

DO'A3' - Search string template

DO'B1' - Search string DO

DO'81' - Search concatenation of country code and certificate serial number: xx xx .. xx xx

Le: 00 (Short Le)

Response: DO'7F76' - Record handling DO

DO'51' - EF.Certificates short identifier '1A'

DO'02' - contains matching record number

Data	SW1-SW2
7F 76 06 51 01 1A 02 01 01	90 00

or warning code 62 82 if no record matches the search criteria:

SW1-SW2
62 82

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If an EF.Certificate record matches the search criteria, the IS can optionally use the returned record number ('01') in a READ RECORD command to check whether the certificate is the correct one. If no EF.Certificate record matches the search criteria, the IS writes the certificate into EF.Certificates using the APPEND RECORD command in step 2) and finally writes the entry record using step 3).

2) APPEND RECORD command writing Certificate

IS writes LDS2-TS Signer certificate into EF.Certificates. The following command can be used for writing certificates:

CLA	INS	P1	P2	Lc	Data	Le
00	E2	00	D0	00 XX XX	5F3A Len {certificate serial number} 72 Len {X.509 certificate}"	Absent

CLA: Interindustry class / no secure messaging

INS: APPEND RECORD

P1: 00 (any other value is invalid)

P2: short EF identifier (=0x1A)

Lc: Record length (Extended Lc)

Data: Record data

Response: success or error code

SW1-SW2
90 00

3) APPEND RECORD command writing Travel Record

IS generates Travel Record using reference to LDS2-TS Signer certificate and writes it into EF.EntryRecords using the following command:

CLA	INS	P1	P2	Lc	Data	Le
00	E2	00	08	00 XX XX	5F44 Len {destination state} 73 Len {Entry travel record} 5F37 Len {Signature} 5F38 Len {Cert Ref}	Absent

CLA: Interindustry class / no secure messaging

INS: APPEND RECORD

P1: 00 (any other value is invalid)

P2: short EF identifier (=0x01)

Lc: Record length (Extended Lc)

Data: Record data

Response: success or error code

SW1-SW2
9000